

Cove Full Dissertation Bilingual Review

Cove 论文全文中英逐段通读版

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English

Project Submission Pro-Forma Student name: [Student name]

中文

项目提交 申请申请 学生名字: [学生名字]

English

Student ID: [Student ID]

中文

学生身份证: [学生身份证]

English

I wish the dissertation to be considered for the course (select one only):

中文

我希望论文被考虑在课程中 (只选择一个):

English

☐ MSc in Applied Artificial Intelligence I confirm that I have included in my dissertation:

中文

应用人工智能士学位 我确认我在论文中包括:

English

☐ An abstract of the work completed ☐ A declaration of my contribution to the work and its suitability for the degree ☐ A table of contents ☐ A list of figures and tables (if applicable)

中文

☐ 完成的作品摘要 ☐ 关于我对工作的贡献和其适合程度的声明 ☐ 内容表 ☐ 图表和表表列表 (如有)

English

☐ A glossary of terms (where appropriate)

中文

☐ 词典 (如有必要)

English

☐ A clear statement of my project objectives ☐ A full reference list using the Warwick Harvard style ☐ An appendix containing email confirmation of ethical approval or waiver Ethical Approval Waived.

中文

☐ 根据华威克·哈佛风格的完整参考清单 ☐ 附录中包含电子邮件确认道德认可或放弃道德认可.

English

Signed: Date:

中文

签名: 日期:

English

Design and Evaluation of a Lightweight Complete-Outfit Recommendation System Using Frozen SigLIP Representations, Task-Specific Compatibility Learning and Constrained Candidate Search by [Student name], [qualifications obtained]

中文

设计和评估使用冻结的SigLIP表示, 任务特定兼容性学习和[学生名字], [获得资格]的限制候选搜索的轻量级完整穿搭推荐系统

English

Dissertation submitted in partial fulfilment for the Degree of Master of Science in Applied Artificial Intelligence WMG University of Warwick Submitted [Month Year]

中文

部分完成申请应用人工智能科学士学位的论文 WMG Warwick大学提交 [月年]

Declaration

声明

English

I have read and understood the rules on cheating, plagiarism and appropriate referencing as outlined in my handbook and I declare that the work contained in this assignment is my own, unless otherwise acknowledged.

中文

我已经阅读并理解了我手册中所述的欺诈, 抄袭和适当引用规则,

English

No substantial part of the work submitted here has also been submitted by me in other assessments for this or previous degree courses, and I acknowledge that if this has been done an appropriate reduction in the mark I might otherwise have received will be made.

中文

我在本课程或之前的学位课程的其他评估中也没有提交过任何本质性的工作, 我承认如果这样做, 我可能会得到的分数将会适当减少.

English

Project definition for my degree: [Insert the exact wording from the Specific Course Regulations for MSc Applied Artificial Intelligence.]

中文

项目定义: [从士应用人工智能专业课程规则中插入精确的措辞.]

English

Relationship of this project to the degree definition: [To be written after the exact course-regulation wording has been verified. This declaration must remain within one page.]

中文

项目与学位定义的关系: [经过确切课程规范词已被验证后写. 本声明必须保持在一个页面内.]

Abstract

摘要

English

Complete-outfit recommendation requires a system to model relationships among garments while respecting practical constraints and the item already selected by a user. This study evaluates whether a lightweight compatibility model can support that task without training a large visual encoder or restricting the output to one retailer's catalogue. Frozen 768-dimensional SigLIP image representations were combined with a 379,265-parameter compatibility head and trained on the published Polyvore Outfits disjoint compatibility split. The model was compared with non-learned, linear, re-current, type-aware and attention-based controlled baselines over three random seeds. Evaluation included classification, corrected FITB ranking, calibration, structural ablation, imbalance, overfitting, runtime, bias and domain shift.

中文

完整穿搭推荐需要一个系统来建模服装之间的关系,同时尊重实际的限制和用户已经选择的项目。本研究评估了轻量级兼容性模型是否可以支持该任务,而不需要训练一个大型视觉编码器或限制输出到一个零售商的目录。冻结的768维SigLIP图像表示与379,265参数兼容性头结合,并在发表的Polyvore 穿搭分离兼容性分区上训练。该模型与非学习,线性,循环,基于类型感知和注意力控制的基线进行了比较。评估包括分类,修正FITB排名,校准率,结构消融,不平衡,过拟合,运行时间,偏见和域转移。

English

The selected compatibility model achieved a held-out ROC-AUC of 0.8583 (95% bootstrap confidence interval 0.8524-0.8641) and 60.25% FITB accuracy (95% confidence interval 58.73-61.75). The three-seed mean FITB result was 59.94%. Removing the pairwise relationship module reduced mean FITB by 10.28 percentage points, with a paired 95% interval of 8.96- 11.57 points; the smaller gain from slot embeddings was inconclusive. Training AUC was 0.9699, but validation and test AUC differed by only 0.00145 after validation-based early stopping.

中文

选择的兼容性模型实现了0.8583 (95%启动位置保证间隔0.8524-0.8641) 和60.25%FITB (95%保证间隔58.73-61.75)。三种平均FITB结果为59.94%。删除双向关系模块降低了平均FITB10.28个百分点,对95个百分点的间隔为8.96-11.57分;较小的利从位置内置是不确定的。训练AUC为0.9699,但验证和测试AUC在验证基础上的早期停止后只有0.00145分不同。

English

The model was integrated into an abstract, product-independent candidate space with weather and clothing-range constraints. Exact search exposed substantial loss from the previous quota path, which recovered the exact top result in 27.78% of 18 controlled queries. The findings support the feasibility of a lightweight and auditable system, but not universal preference or commercial generalisability: official FITB coverage was 29.50%, pure-menswear evidence was sparse, catalogue domain shift was strong, and no participant study was conducted.

中文

该模型被集成到一个抽象的,产品独立的候选空间,有天气和服装范围的限制。精确搜索暴露了之前的配额路径的大幅损失,该路径在18个受控查询中恢复了27.78%的精确最高结果。研究结果支持轻量级和可审计的系统的可行性,但不是普遍的偏好或商业通用性:官方FITB覆盖率为29.50%,纯人服证据稀缺,目录域转移强,并没有参与者研究进行。

English

Keywords: fashion recommendation; outfit compatibility; SigLIP; fill-in-the-blank; constrained search; responsible artificial intelligence

中文

关键词:时尚推荐;穿搭兼容性;SigLIP;填空;限制搜索;负责任的人工智能

Acknowledgements

感谢

English

[Optional. To be completed by the student.]

中文

学生必须完成.

Chapter 1: Introduction

第一个章: 介绍

1.1 Background and context

1.1 背景和背景

English

Digital fashion recommendation covers several related but distinct tasks, including the retrieval of visually similar products, the identification of substitutes, individual-item recommendation and the assessment of whether a set of garments forms a coherent outfit. Visual resemblance alone is insufficient for the final task. Two shirts may be substitutes because they serve a similar role, whereas a shirt, trousers and shoes may be complements because their different roles form a plausible ensemble. Early image-based recommendation research formalised this distinction between visual similarity, substitution and complementarity (McAuley et al., 2015). A modern review similarly characterises fashion recommendation as a domain in which visual, semantic, contextual and user-related signals interact, while identifying evaluation, explanation and bias as continuing challenges (Deldjoo et al., 2023).

中文

数字时尚推荐涵盖了几个相关但不同的任务, 包括检索视觉相似的产品, 确定替代品, 个人物品推荐和评估一组服装是否形成一致的服装。仅仅视觉相似性不足以完成最终任务。两个衫可能是替代品, 因为它们发挥着类似的作用, 而衫, 子和鞋子可能是补充品, 因为它们的不同作用构成可信的组合。早期基于图像的推荐研究正式化了视觉相似性, 替代性和补充性之间的区别 (麦克利等人, 2015)。现代的综述也将时尚推荐描述为视觉, 语义, 文本和用户相关的信号相互作用的领域, 同时将评估, 解释和偏见视为持续的挑战 (Deldjoo等人, 2023)。(原文引用: (McAuley et al., 2015); (Deldjoo et al., 2023)。)

English

Outfit compatibility research has therefore moved beyond manually specified colour or category rules towards learned relationships among items. A bidirectional long short-term memory model represented outfits as ordered garment sequences and learned their compatibility (Han et al., 2017). Type-aware embedding spaces subsequently distinguished similarity from compatibility across different item relationships (Vasileva et al., 2018). OutfitTransformer then used task-specific tokens and self-attention to model outfit-level relationships for compatibility prediction and complementary-item retrieval (Sarkar et al., 2023). These approaches demonstrate that complete-outfit recommendation is relational: a candidate should be assessed in the context of the other garments rather than scored as an isolated item. They also demonstrate that architecture, category representation, negative construction and evaluation task affect what a reported score means.

中文

因此, 穿搭兼容性研究已经超越了手动指定的颜色或类别规则, 转向对物品之间的学习关系。一个双向长期内存模型代表服装作为订单的服装序列, 并学习了它们的兼容性 (Han et al., 2017)。类型感知的嵌入空间随后区分了不同物品关系之间的兼容性 (Vasileva et al., 2018)。OutfitTransformer随后使用了任务特定的代币和自我关注, 以模型服装层面的关系来预测兼容性和补充物品检索 (Sarkar et al., 2023)。这些方法表明, 完整穿搭推荐是关系性的: 应在其他服装的背景下评估候选, 而不是作为一个孤立的项目。它们还证明, 建筑, 类别表示, 负面构建和评估任务影响报告的分数的意味着什么。

English

Large-scale vision-language pre-training provides a potential representation layer for smaller downstream studies. SigLIP uses independent sigmoid losses over image-text pairs instead of requiring softmax normalisation across a global batch, and was evaluated as a transferable vision-language representation by Zhai et al. (2023). A frozen SigLIP image encoder can therefore supply reusable features while a smaller model learns the project-specific relationship among garments. Freezing does not make the representation neutral or guarantee transfer to fashion data; it instead re-duces the number of trainable parameters and makes controlled comparisons feasible within the resources of an MSc project.

中文

大规模视觉-语言预训为较小下游研究提供了潜在的表示层。SigLIP使用独立的sigmoid损失对图像文本对相比,而不是需要在全局批量中软max正常化,并被Zhai等评为可转移的视觉-语言表示(2023)。因此,一个冻结的SigLIP图像编码器可以提供可重复使用的功能,而一个较小的模型则学习服装之间的项目特定关系。冷不会使表示中立或保证转移到时尚数据;它反而减少了可训练的参数数数数数,并使控制的比较在MSc项目的资源内可行。

1.2 Problem definition and motivation

1.2 问题定义和动机

English

This dissertation studies a practical complete-outfit task. A user supplies one garment, and the system recommends abstract garment types and colours for the remaining supported positions. The prototype, named Cove, uses four logical slots: an inner top, an optional outer layer, a bottom and shoes. The selected garment remains fixed. Weather conditions, selected style and the user-selected clothing range constrain the candidate space, while alternatives allow the user to retain control over the final choice.

中文

本文研究了一个实用完整服装任务。用户提供一个服装,系统推荐对其余支持位置进行抽象服装类型和颜色。原型名为Cove使用四个逻辑位置:内顶,可选外层,底部和鞋子。所选的服装保持固定。天气条件,所选的风格和用户选择的服装范围约束了候选空间,而替代方案允许用户控制最终选择。

English

The learned problem is complete-outfit compatibility scoring followed by candidate ranking. This is separated from candidate feasibility. For example, weather logic may suppress an outer layer in hot conditions, and slot rules prevent two bottoms from occupying the same outfit. Such constraints should not be inferred indirectly from a compatibility score. The deployed system therefore combines a learned relational score with explicit constraints and a deterministic fallback.

中文

学习的问题是完整穿搭兼容性分数,然后是候选排名。这与候选可行性分开。例如,天气逻辑可能在炎热条件下抑制一个外部层,并且位置规则可以防止两个底部占用相同的服装。这些限制不应该间接地从兼容性分数推断。因此,部署的系统结合了学习的关系分数与明确的限制和确定性倒退。

English

The candidate representation is also deliberately product-independent. Recommendations are expressed as abstract garment states rather than as Zara or other branded products. The states are derived from secondary Polyvore training data and described by slot, broad type, colour, style and clothing range. This design reflects the intended interaction: a recommendation can guide the use of an existing wardrobe or a later purchase without claiming that one retailer contains the only suitable solution. Product-independent candidate generation also creates a search problem because compatibility is a non-additive score over complete combinations. The study consequently examines whether bounded exact search produces materially different results from an earlier quota-limited retrieval path.

中文

候选表示也故意与具体产品无关。推荐以抽象服装状态而不是Zara或其他品牌产品表达。这些状态来自二手 Polyvore 训练数据，并由位置，广泛类型，颜色，风格和服装范围描述。这种设计反映了密集的互动：推荐可以指导使用现有衣柜或后续购买，而不声称一个零售商包含唯一合适的解决方案。产品独立的候选生成也会造成搜索问题，因为兼容性是完全组合的非添加分数。因此，该研究检查了是否有限制的精确搜索与以前的限量搜索路径产生了显著不同的结果。

English

The motivation is both methodological and practical. A large end-to-end model would be difficult to train, compare and audit with the available time and computing resources. A rules-only system is transparent but cannot learn the relationships observed in outfits. The adopted middle position retains a frozen visual representation, trains a comparatively small compatibility head and evaluates the learned component separately from the contextual rules and search procedure.

中文

动机既有方法性又有实用性。一个大型端到端模型将难以与可用的时间和计算资源进行培训，比较和审计。一个仅有规则的系统是透明的，但无法学习服装中所需的关系。采用的中位位置保留了冻结的视觉表示，培训了相对较小的兼容性头部，并将学习的组件与文本规则和搜索程序分开来评估。

English

Figure 1.1 summarises the system boundary from user and context inputs to the functions that produce and preserve recommendations. It also distinguishes learned compatibility ranking from recognition, explicit constraints and wardrobe management.

中文

图1.1概述了系统界限，从用户和背景输入到产生和保存推荐的功能。它还区分了学习的兼容性排名与识别，明确的限制和衣柜管理。

English

Cove is therefore treated as a context-aware complete-outfit recommender system. Search is the mechanism used to explore and rank feasible candidate combinations, rather than a separate product-search claim.

中文

因此，Cove被视为一个意识到环境的完整穿搭推荐系统。搜索是探索和排名可行的候选组合的机制，而不是单独的产品搜索。

English

User and context inputs System functions User-facing outputs Selected garment Garment recognition Primary recommendation image or wardrobe item and feature extraction and diverse alternative City weather, style Candidate generation Model-ranked colour and clothing range and constraint filtering alternatives by position Local wardrobe, saved Complete-outfit Weather-aware clothing pieces and outfits compatibility ranking cue and masked absence Wardrobe storage Saved garment or and outfit editing four-position outfit Figure 1.1: Functional scope of the Cove complete-outfit recommendation system

中文

用户和背景输入系统功能 用户面向输出 选择服装 服装识别 首要推荐图像或衣柜物品和功能提取和多种替代 城市天气, 风格 候选代 按位置排名的模型颜色和服装范围和限制过替代品 地方衣柜, 保存 完整服装 时气意识 服装 件件和穿搭兼容性 排名提示和隐藏缺席 衣柜存储 保存服装或服装编辑四位服装图1.1: Cove完整穿搭推荐系统的功能范围

1.3 Research gap

1.3 研究差距

English

The literature establishes that recurrent, type-aware and attention-based models can learn fashion compatibility (Han et al., 2017; Vasileva et al., 2018; Sarkar et al., 2023). It does not follow that the most complex published architecture is automatically the most appropriate model for a smaller, processed dataset, a frozen visual representation and a resource-constrained application. Reported results are coupled to dataset splits, negative construction, candidate availability and evaluation protocol (Herlocker et al., 2004; Vasileva et al., 2018). The broader fashion-recommender literature also cautions that offline accuracy alone does not resolve questions of explanation, bias or operational use (Deldjoo et al., 2023).

中文

文献指出, 复制型, 意识型和关注型号可以学习时尚兼容性 (Han et al., 2017; Vasileva et al., 2018; Sarkar et al., 2023)。这并没有说明, 最复杂的发表架构是自动地适合较小, 处理的数据集, 冻结的视觉表示和资源有限的应用的模型。报告的结果与数据集分区, 负面构造, 候选可用性和评估协议相结合 (Herlocker et al., 2004; Vasileva et al., 2018)。更多的时尚推文献还警告说, 离线准确性本身并不能解决解释, 偏见或运用问题 (Deldjoo 等人, 2023)。(原文引用: (Deldjoo et al., 2023)。)

English

The gap addressed here is therefore bounded and empirical rather than a claim that outfit recommendation itself is novel. First, it is uncertain whether a small relational head on frozen vision-language features can outperform non-learned and controlled learned alternatives under one consistent pipeline. Second, aggregate compatibility classification does not establish candidate-completion quality, calibration or generalisation; corrected FITB, ablation, repeated seeds and overfitting evidence are also required. Third, a model score does not by itself specify how a practical system should explore a large abstract candidate space under weather and clothing-range constraints. The project addresses these questions jointly while keeping model evaluation distinct from claims about human taste or satisfaction.

中文

因此, 所述的差距更为有限, 更是经验性的, 而不是说穿搭推荐本身是新鲜的。首先, 不确定是否在一个一致的管道下, 冻结的视觉-语言特征上的小关系头可以超过未经学习和控制的学习替代品。第二, 综合兼容性分类并不确定候选完成质量, 校准或泛化; 还需要纠正FITB, 废除, 重复种子和过度证明。第三, 模型分数本身不说明实用系统应该如何在天气和服装范围的约束下探索大型抽象候选空间。该项目共同解决这些问题, 同时将模型评估与关于人类品味或满意度的说法分开。

1.4 Research question

1.4 研究问题

English

The primary research question is:

中文

首先, 研究问题是:

English

To what extent can a lightweight outfit-compatibility model built on frozen vision-language embeddings, combined with exact constrained search over abstract garment states, provide accurate, computationally practical and auditable complete-outfit recommendations using secondary fashion data?

中文

基于冻结的视觉-语言嵌入式的轻量级穿搭兼容性模型, 以及对抽象服装状态进行精确的有限搜索, 可以在使用二级时尚数据提供准确, 计算实用和可审计的完整穿搭推荐的程度是多少?

English

Three sub-questions operationalise the model and system components:

中文

模型和系统组件的运行部分包括三个问题:

English

RQ1. How does the proposed compatibility model compare with non-learned and controlled learned baselines under consistent data and evaluation protocols?

中文

RQ1. 根据一致的数据和评估协议, 拟议的兼容性模型与未经学习和受控制的学习基线如何相比?

English

RQ2. Which architectural and training choices are supported by compatibility classification, corrected FITB, calibration, ablation and generalisation evidence?

中文

RQ2. 符合性分类, 修正FITB, 校准, 除和通用证据支持哪些建筑和培训选择?

English

RQ3. What completeness, latency and control trade-offs arise when quota-limited retrieval is replaced by exact constrained search over product-independent abstract garment states?

中文

RQ3. 产品独立的抽象服装国家被精确限制搜索取代时, 会产生什么样的完整性, 延迟和控制权的折衷?

English

The wording to what extent requires negative as well as positive evidence. A high AUC is treated as evidence of discrimination, not as the percentage of recommendations that users will like. Model ranking, uncertainty, runtime, data coverage, bias and domain shift therefore contribute to the answer.

中文

对于使用者来说, 高AUC是歧视的证据, 而不是用户所喜欢的推荐的百分比。

1.5 Research objectives

1.5 研究目标

English

The research question is addressed through the following objectives:

中文

研究问题通过以下目标解决:

English

O1. Critically review fashion compatibility modelling, vision-language representation learning, candidate retrieval and responsible recommendation evaluation.

中文

审核时尚兼容性建模, 视觉-语言代表性学习, 候选检索和负责任的推评估.

English

O2. Construct an ethically bounded and reproducible secondary-data pipeline consistent with the approved research scope.

中文

建立一个符合批准的研究范围的伦理上有界限且可复制的二次数据管道.

English

O3. Develop a lightweight compatibility model using frozen SigLIP embeddings and the published Polyvore disjoint compatibility labels.

中文

开发一个轻量级的兼容性模型, 使用冻结的SigLIP嵌入式和已发布的Polyvore分离兼容性标签.

English

O4. Compare non-learned, linear, recurrent, type-aware and attention-based controlled baselines under consistent splits and three random seeds.

中文

根据一致的分区和三个随机种子, 比较非学习, 线性, 复发, 类型感知和基于注意力的控制基线.

English

O5. Evaluate discrimination, completion ranking, calibration, structural ablation, imbalance, overfitting, runtime, bias and domain shift.

中文

评估歧视, 完成排名, 校准, 结构性废除, 失衡, 过拟合, 运行时间, 偏见和领域转移.

English

O6. Integrate the validated model into a product-independent abstract recommendation system with exact search, weather constraints, clothing-range selection and deterministic fallback.

中文

合并验证的模型, 并将其集成到产品独立的抽象推荐系统中, 包括精确搜索, 天气限制, 服装范围选择和确定性反弹.

English

O7. Critically assess the strength, limitations and practical implications of the evidence without overstating generalisation or user benefit.

中文

批判性评估证据的实力, 局限性和实际影响, 而不夸大通用或用户利益。

1.6 Scope and boundaries

1.6 范围和界限

English

The investigation uses secondary datasets and does not conduct interviews, questionnaires or experiments with human participants. Ethical approval was waived; the confirmation and required ethics-training evidence are reserved for the appendices. The separately published Zara dataset retained locally is used for recognition and feature-domain diagnostics, not as the sole recommendation source.

中文

调查使用二次数据集, 并没有与人类参与者进行采访, 问卷或实验。伦理批准被放弃; 确认和必要的伦理训练证据保留在附件中。地方保存的单独发布的Zara数据集用于识别和特征领域诊断, 而不是唯一的推荐来源。

English

The study does not claim to learn an objective or universal definition of fashion compatibility. Demographic attributes are not collected, so demographic fairness and user satisfaction cannot be measured (Mehrabi et al., 2021; Selbst et al., 2019). The interface asks the user to choose a menswear or womenswear candidate range; it does not infer gender from a person or garment image (Keyes, 2018). Sparse pure-menswear training data preclude a claim that both ranges have equal model reliability. Long-term personalisation from clicks is also outside scope because defensible online learning would require exposure logging, feedback-loop controls and a separate evaluation design (Chaney et al., 2018). Local wardrobe data do not retrain the compatibility model.

中文

该研究并没有声称学习时尚兼容性的客观或普遍定义。人口特征并没有收集, 因此人口公平性和用户满意度无法测量 (Mehrabi et al., 2021; Selbst et al., 2019)。界面要求用户选择男装或女装候选系列; 它不根据一个人或服装图像推断性别 (Keyes, 2018)。稀少的纯男装培训数据排除了说两种范围都有相同的模型可靠性。从点击中进行长期定制也是无法实现的, 因为可辩护的在线学习需要暴露记录, 反馈循环控制和单独的评估设计 (Chaney et al., 2018)。地方衣柜数据不重新训练兼容性模型。

English

The evaluated representation is limited to an inner top, optional outer layer, bottom and shoes. Dresses, jumpsuits and other one-piece garments would require an explicit multi-slot occupancy representation and separate training and evaluation. Accessories and culturally specific garment structures are not modelled comprehensively. Official FITB reporting is restricted to questions representable by these four slots and available frozen embeddings; coverage is reported so that the filtered result is not presented as a full official benchmark score.

中文

评估的代表性仅限于内部, 外层, 底部和鞋子。服装, 跳服和其他单件穿搭需要明确的多个机场占用表现和单独培训和评估。配件和文化特异性服装结构没有全面的模型。官方FITB报告仅限于可通过这些四个机场提供结嵌入式的问题; 报道覆盖性, 以便未将过结果作为完整的官方基准分数呈现。

English

The work evaluates offline compatibility and system behaviour, not commercial conversion or human preference. Earlier engineering models use different candidate or data protocols and are not treated as causal evidence that one development stage improves another. Controlled claims are based on same-split baselines, architecture comparisons, structural ablations and the exact-versus-quota search experiment.

中文

该研究评估了离线兼容性和系统行为, 而不是商业转换或人类偏好。早期的工程模型使用不同的候选或数据协议, 并没有被视为一个开发阶段改善另一个的因果证据。控制的索赔基于相同的分区基线, 建筑比较, 结构性消融和精确对比配额搜索实验。

1.7 Contributions

1.7 贡献

English

Within these boundaries, the dissertation makes four contributions. First, it provides a reproducible lightweight compatibility pipeline that separates a frozen visual representation, a learned relational head and explicit contextual constraints. Second, it provides a controlled evaluation comprising non-learned and learned baselines, three seeds, corrected set-identifier-based FITB labels, structural ablation, calibration and an explicit overfitting audit. Third, it develops an abstract candidate representation and evaluates bounded exact search against a quota-limited retrieval path using the same deployed objective. Fourth, it demonstrates a local-first prototype in which the user retains the selected garment, chooses the clothing range, receives weather-aware alternatives and can fall back to deterministic rules if model artefacts are un-available.

中文

在这些界限内, 论文做出了四项贡献。第一, 它提供了一个可复制的轻量兼容性管道, 分离了冻结的视觉复现, 学习的关系头和明确的文本限制。第二, 它提供了一个由未学习和学习的基线, 三种种子, 基于设置识别器的修改的FITB标签, 结构性消除, 校准和明确的过度审计组成的控制评估。第三, 它开发了一个抽象的候选表示, 并通过相同的部署目标来评估与一个限量限制的检索路径相对的有限精确搜索。第四, 它展示了一个本地第一种原型, 用户保留所选的服装, 选择服装范围, 接受天气知情的替代方案, 如果模型文物不出现在手中, 则可以回到确定性规则。

English

The contributions are methodological, empirical and artefactual. They do not constitute a new general theory of fashion compatibility, an exact reproduction of every comparison architecture, evidence of user preference, or a claim of state-of-the-art performance on the complete official Polyvore benchmark.

中文

这些贡献是方法, 经验和艺术的。它们不构成一种新的一般的时尚兼容性理论, 对每个比较架构的精确复制, 用户偏好的证据, 或者在完整的官方Polyvore基准上声称最先进的性能。

1.8 Dissertation structure

1.8 论文结构

English

Chapter 2 critically reviews recommender-system foundations, fashion compatibility, vision-language representations, candidate search, evaluation and responsible-system concerns. Chapter 3 specifies the secondary-data pipeline, model, baselines, experimental protocol, metrics, candidate-search system and ethical safeguards. Chapter 4 reports the model and system results without extended interpretation. Chapter 5 analyses model behaviour, architectural evidence, overfitting and search trade-offs. Chapter 6 relates the findings to previous research, answers the research questions and evaluates contribution, validity, practical implications and limitations. Chapter 7 concludes by assessing the objectives and identifying proportionate future work.

中文

第二章批判性综述推荐系统的基础, 时尚兼容性, 视觉-语言的表示, 候选搜索, 评估和负责系统的关注。第三章具体说明了二级数据管道, 模型, 基线, 实验协议, 指标, 候选搜索系统和道德保障措施。第四章报告了模型和系统结果, 没有扩大解释。第5章分析了模型行为, 建筑证据, 过拟合和搜索交易。第6章将发现与以前的研究联系起来, 回答研究问题, 评估了贡献, 有效性, 实用影响和限制。第7章通过评估目标和确定相对的未来工作来结束。(原文数值: 2、3、4。)

1.9 Summary

1.9 总结

English

This chapter has framed complete-outfit recommendation as relational compatibility scoring followed by constrained candidate search. It has defined a bounded empirical gap, one primary research question, three sub-questions and seven objectives. The scope excludes participant claims, online personalisation and unsupported garment structures, while retaining model, search, runtime, bias and reproducibility evidence. Chapter 2 now evaluates the literature from which the representation, compatibility and assessment choices are derived.

中文

本章将完整穿搭推荐设为关系互容性评分, 其次是限制候选搜索。它定义了一个有限的实验差距, 一个主要研究问题, 三个子问题和七个目标。该范围不包括参与者索赔, 在线定制和不支持的服装结构, 同时保留模型, 搜索, 运行时间, 偏见和可复制性证据。第二章现在评估了代表性, 兼容性和评估选择所衍生的文献。(原文数值: 2。)

Chapter 2: Literature Review

第二章 翻译文献

2.1 Introduction

2.1 引言

English

This chapter critically reviews the research underpinning the project's representation, compatibility-modelling, retrieval and evaluation choices. Fashion recommendation includes visual similar-item retrieval, substitute recommendation, personalised rank-ing, outfit compatibility prediction and complementary-item retrieval (Deldjoo et al., 2023; Sarkar et al., 2023). Because these tasks differ in their inputs, supervision and definitions of success, evidence obtained for one cannot automatically support another.

中文

本章批判性地综述了该项目的代表性, 兼容性建模, 检索和评估选择的研究. 时尚推荐包括视觉类似项目检索, 替代推荐, 个性化排名, 穿搭兼容性预测和补充项目检索 (Deldjoo等人, 2023; Sarkar等人, 2023). 由于这些任务在输入, 监督和成功定义方面不同, 因此获得的证据不能自动支持另一个. (原文引用: (Deldjoo et al., 2023; Sarkar et al., 2023).)

English

The review first positions complete-outfit recommendation within recommender systems, then examines the development of fashion-compatibility models and the role of frozen vision-language representations. It subsequently considers learning objectives, evaluation, candidate search and responsible design. The literature is used to derive controlled alternatives, not to assume that the newest or most complex architecture is necessarily appropriate.

中文

综述首先在推荐系统主题中设置了完整穿搭推荐, 然后检查了时尚兼容性模型的发展和冻结的视觉-语言表示的作用. 随后它考虑了学习对象, 评估, 候选搜索和负责任设计. 文献用于推导可控制的替代品, 而不是假设最新或最复杂的架构是必要的.

2.2 Positioning complete-outfit recommendation

2.2 定位完整穿搭推荐

English

Traditional recommender systems often learn from interactions such as ratings, clicks or purchases and ask which item should be presented to a particular user. Evaluation must be connected to the user task, available observations and intended use; otherwise, nominally similar experiments may not be comparable (Herlocker et al., 2004). Fashion recommenders extend this setting with visual, textual, contextual and social information (Deldjoo et al., 2023).

中文

传统的推荐系统通常从评级, 点击或购买等互动中学习, 并询问应该向特定用户展示哪些项目. 评估必须与用户任务, 可用的观察和预期使用有关; 否则, 名义上类似的实验可能无法比较 (Herlocker等人, 2004). 时尚推荐者将这种设置扩展到视觉, 文本, 文本和社会信息 (Deldjoo等人, 2023). (原文引用: (Herlocker et al., 2004); (Deldjoo et al., 2023).)

English

Cove has no longitudinal user-item history and therefore does not claim to learn stable individual preferences or compare collaborative-filtering algorithms. Its learned task is content-based outfit compatibility: estimating whether relationships within a set of garments resemble those observed in the training outfits. At runtime, that score ranks completions around a user-selected garment. Dataset compatibility and personal preference must remain distinct, because following a dataset regularity does not establish what one person will prefer.

中文

科夫没有纵向用户项目历史, 因此不声称学习稳定个人偏好或比较协作过滤算法。其所学习的任务是基于内容的穿搭兼容性: 估计服装组内的关系是否类似于训练服装中观察到的情况。在运行时间, 该分数将围绕用户选择的服装进行排名。数据集兼容性和个人偏好必须保持着区别, 因为遵循数据集的规律性不会确定一个人将喜欢什么。

English

Complete-outfit recommendation is consequently considered at three connected levels. Representation converts garment images into features; relational scoring evaluates garments jointly; decision and interaction filter infeasible candidates, search combinations and expose alternatives. Secondary outfit data can evaluate the first two levels. The third can be tested for deterministic behaviour, completeness and runtime, but not for satisfaction without human evidence. This separation prevents the compatibility classifier from being treated as the entire recommender system.

中文

后者可以评估前两级。第三级可以测试确定性行为, 完整性和运行时间, 但不能测试人体证据。这种分离防止兼容分类器被视为整个推荐系统。

2.3 Development of fashion-compatibility models

2.3 时尚兼容性模型的开发

English

Early image-based recommendation distinguished visual similarity from complementarity. Items serving the same role may be substitutes, whereas items from different roles may be complementary despite being visually dissimilar (McAuley et al., 2015). Pairwise metric learning operationalises this distinction by learning spaces in which compatible pairs are close. Such models support efficient nearest-neighbour retrieval, but a complete outfit contains several interacting pairs. Summing local scores assumes that pairwise compatibility is sufficient to represent the set.

中文

早期基于图像的推荐区分了视觉相似性与补充性。服务于相同角色的项目可能是替代品, 而不同角色的项目可能是相辅相似的, 尽管视觉不同 (McAuley等人, 2015)。对对的测量学习通过学习空间实现这种区别, 其中相容的对是接近的。此类模型支持高效的近邻检索, 但完整的服装包含几个互动的对。将本地分数作为对对相容性足以代表集合。(原文引用: (McAuley et al., 2015)。)

English

Han et al. (2017) represented outfits as ordered garment sequences and used a bi-directional long short-term memory network. This aggregates information from garments on both sides of an item, although outfits have no natural linguistic order and must be ordered through garment categories. The method therefore provides a useful structured baseline while retaining possible sensitivity to that imposed sequence.

中文

汉等人 (2017) 代表服装作为订单服装序列, 并使用双向长期内存网络。这将从物件的两侧服装中收集信息, 尽管服装没有自然语言顺序, 并且必须通过服装类别进行订单。因此, 该方法提供了一个有用的结构基线, 同时保持对该被强加的序列的可能敏感性。

English

Vasileva et al. (2018) argued that compatibility depends on item type and proposed type-aware embedding spaces. A top-shoe relationship need not use the same transformation as a top-top relationship, making category or slot identity part of the learned relation rather than descriptive metadata. This supports testing slot-pair-specific projections and explicit slot representations.

中文

瓦西莱娃等人 (2018) 认为兼容性取决于项目类型和拟议的类型感知嵌入空间。顶鞋关系不需要使用与顶顶关系相同的转型, 使类别或位置身份成为学习关系的一部分而不是描述性元数据。这支持测试位置对特定的投影和明确的位置表示。

English

Context-aware graph modelling provides another response to isolated pair relations. Cucurull et al. (2019) generated product embeddings conditioned on known compatible context and reported stronger compatibility prediction when more context was available. The finding supports the view that compatibility is conditional rather than intrinsic to one item. Graph construction, however, requires available nodes and edges that may not exist for a newly uploaded garment.

中文

背景意识图建模提供了对孤立对关系的另一种反应。Cucurull et al. (2019) 产生了基于已知兼容的背景条件的产品嵌入, 并在更多的背景可用的时报告了更强大的兼容性预测。该发现支持了兼容性是一个项目的条件而不是固有的观点。

English

Attention allows each garment representation to interact with every other garment without requiring a fixed sequence. OutfitTransformer uses task-specific tokens and self-attention for outfit compatibility and complementary-item retrieval (Sarkar et al., 2023). Treating an outfit as an unordered set is appropriate because rearranging the same garments should not change compatibility. Attention offers flexible higher-order interaction but also adds capacity. Results obtained under OutfitTransformer's original protocol do not establish that equal capacity is required with frozen inputs and a smaller processed dataset.

中文

注意力允许每一件服装的表示与其他服装交互, 而不需要固定序列。OutfitTransformer使用任务特定的代币和自我注意力来实现穿搭兼容性和补充物品检索 (Sarkar等, 2023)。将服装视为一个无序的集合是合适的, 因为重新安排相同的服装不应该改变兼容性。注意力提供灵活的更高级交互, 但也增加了容量。根据OutfitTransformer的原始协议获得的结果并不证明, 冷输入和更小的处理数据集需要相同的容量。(原文引用: (Sarkar et al., 2023)。)

English

These studies support several modelling assumptions rather than one universal architecture. Pairwise models prioritise local relations and efficiency; recurrent models learn structured aggregation; type-aware models encode category-dependent relations; and attention learns set-level interactions (Han et al., 2017; Vasileva et al., 2018; Sarkar et al., 2023). Their differences motivate controlled comparisons using the same features, splits and training conditions.

中文

这些研究支持了几个建模假设, 而不是一个普遍的建筑结构。双向模型优先考虑本地关系和效率; 复发模型学习结构化集成; 类型感知模型编码类别依赖关系; 注意力学习设置层次的相互作用 (Han et al., 2017; Vasileva et al., 2018; Sarkar et al., 2023)。它们的差异激励使用相同的特征, 分区和培训条件进行控制比较。

2.4 Polyvore data and label construction

2.4 多食用数据和标签结构

English

Polyvore datasets convert outfits assembled by platform users into observable item sets and are consequently practical secondary data for compatibility research (Han et al., 2017; Vasileva et al., 2018). A positive example records an outfit retained in the dataset; it is not evidence of an objective or universally shared aesthetic judgement.

中文

多体数据集将平台用户组装的服装转化为可观测的项目集, 因此是兼容性研究的实际二次数据 (Han et al., 2017; Vasileva et al., 2018). 一个积极的例子记录了数据集中保留的服装; 它不是对客观或普遍共享的审美判断的证据。

English

Negative outfits are commonly constructed by replacing an item in a positive outfit. Same-type replacement is harder than arbitrary-category replacement because the negative remains structurally plausible (Vasileva et al., 2018). Since replacements are generated rather than independently judged incompatible, they may include false negatives. This risk follows from the published construction procedure; it does not establish that a known proportion of labels is incorrect.

中文

消极服装通常是通过在正面服装中取代物品而构建的。同类型的替换比任意类型的替换更难, 因为负面仍然是结构上可行的 (Vasileva et al., 2018). 由于替换产品是由独立判断不兼容而不是产生, 它们可能包括虚假消极。这种风险来自于已发表的建筑程序; 它并没有确定已知的标签比例是不正确的。

English

Splitting the data changes the generalisation question. If a product appears in both training and testing outfits, a model may benefit from recognising an observed item. The Polyvore Outfits disjoint split prevents item overlap across its principal divisions and gives a more demanding test on unseen garments (Vasileva et al., 2018). It does not remove shared photography, category distributions, platform culture or labelling assumptions.

中文

分析数据的分化改变了泛化问题。如果一个产品出现在训练和测试服装中, 模型可能会从识别观察到的物品中受益。多服装分离防止物品在其主要部门中重叠, 并对未见的服装进行更苛刻的测试 (Vasileva et al., 2018). 它不删除共享摄影, 类别分布, 平台文化或标签假设。

English

In the common fill-in-the-blank (FITB) task, one garment is removed and the model selects the correct completion from four candidates (Han et al., 2017; Vasileva et al., 2018). FITB evaluates relative ranking within a small set and is closer to recommendation than binary classification, but remains smaller than full-catalogue retrieval. Out-fitTransformer evaluates compatibility prediction and large-scale retrieval separately, reinforcing that FITB does not establish scalable retrieval quality (Sarkar et al., 2023).

中文

在常见的填空 (FITB) 任务 (FITB) 中, 一个服装被删除, 模型从四个候选中选择了正确的完成 (Han et al., 2017; Vasileva et al., 2018). FITB评估了在一个小组内相对排名, 并且更接近于推荐, 而不是二进制分类, 但仍然比完整目录检索小。Out-fitTransformer单独评估了兼容性预测和大规模检索, 加强了FITB不具有可扩展的检索质量 (Sarkar et al., 2023)。

English

Polyvore is therefore suitable for controlled training and offline evaluation, but cannot represent all clothing practices. Filtering official questions to four supported slots introduces an additional coverage boundary. The retained fraction must accompany the subset result so that it is not presented as the complete official benchmark.

中文

由于使用多种类型的服装, 它可以用于控制训练和离线评估, 但不能代表所有服装实践。过官方问题到四个支持的位置引入了额外的覆盖范围。保留的分数必须随着子集结果, 以便它不被呈现为完整的官方基准。

2.5 Frozen vision-language representations

2.5 视觉-语言的结表现

English

Earlier compatibility systems commonly used convolutional encoders pretrained on ImageNet and adapted to fashion data (Vasileva et al., 2018). Vision-language models instead learn from large collections of image-text pairs. CLIP demonstrated transfer-able visual representations and zero-shot recognition across numerous downstream datasets (Radford et al., 2021).

中文

之前的兼容性系统通常使用了Convolutional encoders pretrained on ImageNet, 并适应了时尚数据 (Vasileva等人, 2018)。视觉-语言模型相反, 从图像和文本对的大量集合中学习。CLIP在许多下游数据集中展示了可转移的视觉表示和零截图识别 (Radford等人, 2021)。(原文引用: (Vasileva et al., 2018); (Radford et al., 2021)。)

English

SigLIP belongs to this family but replaces a global softmax contrastive objective with independent sigmoid losses over image-text pairs (Zhai et al., 2023). It was not pretrained for outfit compatibility; its image encoder instead supplies a general visual representation from which a smaller downstream model can attempt to learn garment relationships.

中文

SigLIP属于这个家庭, 但以独立的图像文字对象的sigmoid损失取代了全局softmax对比目标 (Zhai et al., 2023)。它并没有为穿搭兼容性预训练; 其图像编码器相反提供了一般的视觉表示, 从中一个较小的下游模型可以尝试学习服装关系。

English

Freezing the encoder converts each image into a fixed vector and restricts training to the compatibility head. For this project, that reduces trainable parameters, repeated image-processing cost and representation variation across baselines. It also permits comparisons with identical inputs. These are computational and experimental advantages within the project, not evidence that frozen features are universally preferable.

中文

结编码器将每个图像转换为固定向量, 并限制了训练到兼容性头。对于这个项目, 减少可训练的参数, 重复图像处理成本和基线之间的表示变化。它还允许与相同输入进行比较。这些是项目内的计算和实验性优点, 而不是证明结特征是普遍的优点。

English

The downstream head cannot recover information absent from the frozen vector. Transfer is not guaranteed by source-model scale alone: Kornblith et al. (2019) found that transfer depends on both representation and target task, with limited benefit on some fine-grained datasets. Web-scale pretraining can also transmit biases in the source data (Radford et al., 2021; Mehrabi et al., 2021). Frozen SigLIP is thus a resource-conscious hypothesis to test rather than an assumed fashion-specific solution.

中文

下游头部不能从结的向量中恢复缺失的信息。仅仅通过源模型尺度来保证转移: Kornblith等人(2019)发现, 转移取决于表示和目标任务, 对一些细粒度的数据集具有有限的好处。网络尺度预训也可以传输源数据中的偏见 (Radford等人, 2021; Mehrabi等人, 2021)。因此, 结的SigLIP是一种资源意识的假设来测试而不是一种假设的时尚特定解决方案。(原文引用: (Radford et al., 2021; Mehrabi et al., 2021)。)

2.6 Learning objectives and controlled baselines

2.6 学习目标和控制的基线

English

Compatibility classification and recommendation ranking are related but distinct. Binary cross-entropy trains separation of labelled positive and negative outfits. Its score can order candidates, but the loss does not directly optimise order among competing completions. Pairwise ranking instead encourages a preferred observation to score above an alternative; Bayesian Personalised Ranking is one established formulation for implicit observations (Rendle et al., 2009).

中文

兼容性分类和推排名是相似的, 但不同的。二进制交叉熵训练标记的积极和负面服装的分离。它的分数可以订单候选, 但损失不会直接优化竞争完成之间的顺序。对称排名反而鼓励一个优先观察比一个替代品; 贝耶斯人格化排名是隐含观察的一个确定的公式 (Rendle et al., 2009)。

English

Neither objective is automatically preferable here. Binary labels are directly available in the compatibility rows, whereas ranking requires defensible positive-negative pairs. If a synthetic negative is noisy, a ranking objective may optimise an unreliable comparison. The choice must be evaluated rather than inferred from the fact that runtime recommendation uses ranking.

中文

由于二进制标签在兼容性行中直接可用, 而排名需要可辩护的正负对。如果合成负是噪音, 排名目标可能会优化一个不靠谱的比较。选择必须从运行时间推使用排名的事实上进行评估, 而不是推断。

English

The controlled baselines form a hierarchy. Chance provides a lower bound for four-choice FITB. Non-learned cosine similarity tests whether frozen features already contain useful structure. A linear model over pooled features tests whether learned weighting is sufficient without nonlinear interaction. These controls prevent the downstream architecture from receiving credit for information already present in SigLIP.

中文

控制的基线构成一个层次结构。机会为四种选择的FITB提供了下限。非学习的共数相似性测试是否已经包含有用的结构。对集成的特性进行线性模型测试是否是非线性相互作用的情况下学习的权重足够。这些控制阻止下游架构获得SIGLIP中已经存在的信息的信誉。

English

A recurrent baseline represents the sequence assumption of Han et al. (2017); a type-aware model represents relation-specific transformations from Vasileva et al. (2018); and attention represents set-level interaction following Sarkar et al. (2023). These are controlled approximations, not exact reproductions, which would require the original encoders, preprocessing, objectives and hyperparameters.

中文

复发基线代表了汉等人（2017）的序列假设；一种类型感知模型代表了Vasileva等人（2018）的关系特定变化；以及注意力代表了Sarkar等人（2023）后的设置层次互动。这些是控制式近似，而不是精确的复制，这需要原始编码器，预处理，目标和超参数。

English

Structural ablation addresses component contribution. Removing the pairwise module tests whether explicit item-item interaction adds evidence beyond pooled outfit information. Removing slot embeddings tests whether slot identity adds evidence beyond visual vectors. If removal does not produce a consistent decline across repeated runs, the component's benefit remains uncertain.

中文

结构式除处理组件贡献。删除对模块测试是否明确的项目-项目互动增加除了集成的服装信息之外的证据。删除位置嵌入式测试是否增加位置身份证据后面视觉向量。如果删除不会在重复运行中产生一致的下降，组件的好处仍然不确定。

2.7 Evaluation beyond a single score

2.7 超过一个分数的评估

English

Evaluation should match the task claimed (Herlocker et al., 2004). Compatibility classification, candidate completion, confidence interpretation and application behaviour therefore require separate evidence.

中文

评估应与所要求任务相匹配（Herlocker等人，2004）。因此，兼容性分类，候选完成，信任解释和申请行为需要单独的证据。（原文引用：（Herlocker et al., 2004）。）

English

ROC-AUC measures discrimination across thresholds and can be interpreted as the probability that a randomly selected positive receives a higher score than a randomly selected negative (Fawcett, 2006). It does not measure whether the correct completion ranks first in a candidate set, nor whether a score is a calibrated probability.

中文

ROC-AUC测量在门之间歧视，可以解释为随机选择的正值获得比随机选择的负值更高的分数的概率（Fawcett, 2006）。它不测量是否正确的完成在候选组中排名第一，也不能测量是否一个分数是校准的概率。（原文引用：（Fawcett, 2006）。）

English

FITB evaluates completion more directly. In a balanced four-choice question, chance accuracy is 25%, but performance still depends on candidate and label construction. Correct answers must be matched through the documented dataset relationship rather than assumed to occupy a fixed array position. Retaining question-level outputs makes label correction and paired comparison auditable.

中文

根据FITB的数据库, 测试结果是可查的, 但在一个平衡的四选择问题中, 机会准确度为25%, 但性能仍然取决于候选和标签构建. 正确的答案必须通过记录数据集关系进行匹配, 而不是假设占据固定数组位置. 保留问题级输出使标签纠正和对比可审核.

English

Calibration is separate from discrimination. Neural networks may rank examples correctly while remaining overconfident; temperature scaling was effective in several classification settings examined by Guo et al. (2017). Because a positive scalar temperature does not alter logit order, it can improve confidence interpretation without improving FITB ranking.

中文

校准与歧视分开. 神经网络可以正确排名示例, 但仍然过度自信; 温度扩展在几种分类设置中有效, 由郭等人研究. (2017). 由于正面的扩展温度不会改变逻辑顺序, 它可以改善信心解释, 而不改善FITB排名.

English

Stochastic training variation must also be reported. Parameter initialisation, data order and other benchmark stages can alter an otherwise unchanged experiment (Bouthillier et al., 2021). Cove uses three seeded runs for its principal architecture comparisons. This is a resource-constrained repeated-run design, not a complete characterisation of training variance.

中文

必须报告的还要是: 参数初始化, 数据顺序和其他基准阶段可以改变一个以往未改变的实验 (Bouthillier等人, 2021). Cove用于其主要架构比较的三个种子运行. 这是一个资源限制的重复运行设计, 而不是训练变异的完整特征. (原文引用: (Bouthillier et al., 2021).)

English

Structural ablations are evaluated on the same FITB questions and matched seeds. Pairing preserves the full-ablation relationship on each evaluation unit instead of treating their predictions as independent (Dietterich, 1998). The project's bootstrap averages the three seeded outcomes within each question and then resamples questions, following the bootstrap principle of estimating sampling uncertainty by resampling (Efron and Tibshirani, 1993). Its interval is conditional on those runs and does not represent every possible training seed.

中文

结构性除在同一FITB问题和匹配的种子上进行评估. 对配对保留了每个评估单元的完整除关系, 而不是将其预测视为独立的 (Dietterich, 1998). 该项目起步平均每个问题中的三个种子结果, 然后重新样本问题, 遵循采样不确定性通过重新样本估计的起步原则 (Efron和Tibshirani, 1993). 它的间隔取决于这些运行, 并不代表每一个可能的培训种子. (原文引用: (Efron and Tibshirani, 1993).)

English

Overfitting is examined through training, validation and held-out test behaviour. Validation-based early stopping is intended to stop iterative training when further training improvement is no longer accompanied by better validation performance (Prechelt, 1998). A small validation-test difference supports only the consistency of the selected checkpoint on the current test split, not general reliability. Stronger training than validation and test performance is consistent with training-set overfitting, although data composition can also contribute.

中文

过度穿戴通过训练, 验证和持续测试行为进行检查。基于验证的早期停止旨在停止再过度训练, 当进一步的训练改进不再伴随着更好的验证性能 (Prechelt, 1998)。验证与测试的小差异仅支持当前测试分区的选择检查点的一致性, 而不是一般的可靠性。比验证和测试性能更强的训练与训练集过度穿戴一致, 尽管数据组成也可以有助。

English

No offline metric establishes user benefit. Without participant evidence, claims re-main limited to discrimination, controlled completion ranking, calibration, runtime, coverage and deterministic system behaviour.

中文

没有离线指标确定用户的好处。没有参与者证据, 要求重复仅限于歧视, 控制完成排名, 校准, 运行时间, 覆盖和确定性系统行为。

2.8 Candidate search and explicit constraints

2.8 候选搜索和明确的限制

English

A model scores only candidates that reach it. If a strong candidate is removed by a frequency quota or similarity shortlist, later ranking cannot recover it. Candidate generation is therefore part of recommendation behaviour rather than neutral prepro-cessing.

中文

模型只评分到达到它的候选。如果一个强大的候选被频率配额或相似性选项清单删除, 后来的排名无法恢复它。因此, 候选生成是推行为的一部分, 而不是中立的预处理。

English

Large-scale complementary-item systems commonly use embedding retrieval. Outfit-Transformer creates a target representation and applies nearest-neighbour retrieval (Sarkar et al., 2023). Vector indexes and approximate search make this efficient at very large scale, trading retrieval completeness for speed or memory (Johnson et al., 2021).

中文

大规模的补充物件系统通常使用嵌入式检索。Outfit-Transformer创建目标表示并应用近邻检索 (Sarkar等, 2023)。向量指标和近似搜索使得这在非常大规模上效率, 交易检索完整性速度或内存 (Johnson等, 2021)。(原文引用: (Sarkar et al., 2023); (Johnson et al., 2021)。)

English

Cove instead assigns a non-additive score to a complete combination. An item that is not individually closest to the uploaded garment can still participate in the highest-scoring complete outfit, so independent slot shortlists may not preserve the final optimum. This follows from the deployed scoring function rather than from a claim made by the retrieval literature.

中文

科夫则将一个非添加分数分配给一个完整组合。一个单独不接近上传的服装的项目仍然可以参与最高分数的完整服装, 因此独立的位置短清单可能不会保持最终最佳。这来自于部署的分数函数而不是检索文献的要求。

English

For a bounded four-slot space, batched exhaustive scoring can remain feasible. Here, exact means that every combination surviving the declared filters is evaluated under one deployed objective. It does not mean that the model has identified an objectively best human outfit. A safety limit is required so an oversized search fails explicitly rather than presenting a partial prefix as complete.

中文

对于一个有限的四槽空间, 批量完整的分数仍然可行. 在这里, 精确意味着每个幸存于声明过器的组合都被根据一个部署的目标评估. 这并不意味着模型已经确定了客观最佳的人类服装. 安全限制是必要的, 因此一个超大规模的搜索明确失败, 而不是呈现一个部分预写为完整.

English

Weather suitability, slot occupancy and the user-selected clothing range are feasibility requirements, not aesthetic preferences inferred from Polyvore. The model ranks feasible outfits, while deterministic rules define what is allowed. This keeps model behaviour, search completeness and functional constraints separately auditable.

中文

气候适合性, 空隙占用率和用户选择的服装范围是可行性要求, 而不是从Polyvore推断的审美偏好. 模型排名可行的服装, 而确定性规则定义了允许的. 这使得模型行为, 搜索完整性和功能限制可以单独进行审计.

2.9 Bias, personalisation and responsible design

2.9 偏见, 个性化和负责任设计

English

Bias can enter through data selection, labels, pretrained representations, candidate construction and interaction design, and cannot be reduced to overall accuracy (Mehr-abi et al., 2021). Clothing practice is also cultural, temporal and subjective (Deldjoo et al., 2023).

中文

偏见可以通过数据选择, 标签, 预训练的表现, 候选构建和交互设计进入, 不能被降低到总体准确性 (Mehr-abi等人, 2021). (原文引用: (Mehr-abi et al., 2021); (Deldjoo et al., 2023).)

English

Polyvore reflects a historical platform population and may over-represent some categories, presentation styles and commercial photography. Synthetic same-type negatives add another assumption, while SigLIP carries uncertainty from web-scale pre-training. These sources justify an audit and explicit limitations, but do not permit demographic fairness to be measured without demographic attributes.

中文

聚集体体现了历史平台人口, 可能过度代表一些类别, 演示风格和商业摄影. 合成同类消极性增加了另一个假设, 而SigLIP则带来了网络规模预训练的不确定性. 这些来源证明了审计和明确的限制, 但不允许在没有人口特征的情况下测量人口公平性.

English

Automatic gender inference would add an uncertain identity classification, a practice criticised in human-computer interaction research (Keyes, 2018). Cove asks the user to choose a menswear or womenswear candidate range instead. Those terms remain simplified catalogue filters and are not definitions of the user.

中文

自动性推断将增加一个不确定身份分类, 这是人与计算机互动研究中受到批评的做法 (Keyes, 2018). Cove要求用户选择男性服或女性服候选品. 这些术语仍然是简单的目录过滤器, 而不是用户的定义.

English

Local execution reduces external transmission of wardrobe data but does not resolve feedback bias. Exposure influences which clicks become observable, and simulation evidence shows that algorithmically confounded feedback can increase behavioural homogeneity without increasing utility (Chaney et al., 2018). Defensible long-term self-learning would need exposure logging, exploration, feedback-loop controls and separate evaluation, so it lies outside the frozen study.

中文

局部执行减少了衣柜数据的外部传输, 但并没有解决反偏见。点击的曝光影响可以观察到, 模拟证据表明, 算法混的反可以增加行为均性而不增加实用性 (Chaney等人, 2018)。可防范的长期自我学习需要曝光记录, 探索, 反循环控制和单独的评估, 因此它位于冷研究之外。(原文引用: (Chaney et al., 2018)。)

English

Fairness cannot be detached from social context or treated solely as a property of an isolated component (Selbst et al., 2019). The proportionate response is not to claim that bias has been solved, but to document limitations, avoid identity inference, retain the selected garment, expose alternatives and permit rejection or replacement.

中文

公平不能脱离社会背景, 也不能仅仅被视为孤立组件的属性 (Selbst et al., 2019)。比例的反应不是声称偏见已经解决, 而是要记录限制, 避免认同推断, 保留选择的服装, 暴露替代品, 允许拒绝或更换。

2.10 Synthesis and research framework

2.10 综合和研究框架

English

The literature supports four conclusions. First, compatibility is relational: visual similarity is insufficient and category identity can form part of the relation, although maximum model complexity is not always justified (McAuley et al., 2015; Vasileva et al., 2018; Sarkar et al., 2023). This supports evaluating a lightweight relational head against progressively stronger alternatives.

中文

文献支持四个结论。第一, 兼容性是关系性的: 视觉模拟差异性不足, 类别身份可以成为关系的一部分, 尽管最大模型复杂性并不总是合理 (McAuley等人, 2015; Vasileva等人, 2018; Sarkar等人, 2023)。这支持评估轻量级的关系头部与逐步更强大的替代品。(原文引用: (McAuley et al., 2015; Vasileva et al., 2018; Sarkar et al., 2023)。)

English

Second, frozen vision-language features offer a feasible representation strategy rather than a fashion-specific solution. They reduce training burden while retaining transfer, domain and inherited-bias risks (Radford et al., 2021; Zhai et al., 2023; Kornblith et al., 2019). The downstream head must consequently be compared with non-learned and learned controls.

中文

第二, 结视觉-语言功能提供了可行的代表性策略而不是特定时尚解决方案。它们减少了训练负担, 同时保留了转移, 域和遗传偏见风险 (Radford等, 2021; Zhai等, 2023; Kornblith等, 2019)。因此, 下游头部必须与未学习和学习的控制进行比较。(原文引用: (Radford et al., 2021; Zhai et al., 2023; Kornblith et al., 2019)。)

Third, ROC-AUC, FITB, calibration, repeated runs, ablation, coverage and domain diagnostics answer different questions (Fawcett, 2006; Guo et al., 2017; Bouthillier et al., 2021). None establishes human satisfaction.

第三, ROC-AUC, FITB, 校准, 重复运行, 除, 覆盖和域诊断回答不同的问题 (Fawcett, 2006; Guo et al., 2017; Bouthillier et al., 2021). (原文引用: (Fawcett, 2006; Guo et al., 2017; Bouthillier et al., 2021)).)

Fourth, candidate generation and explicit constraints affect deployed behaviours as well as scoring. Approximate retrieval is appropriate at very large scale, but a bounded non-additive four-slot objective creates a testable case for exact constrained search (Sarkar et al., 2023; Johnson et al., 2021).

[illegible]

The remaining empirical questions are whether a small relational head on frozen SigLIP features can balance discrimination, completion ranking, runtime and audit-ability under one secondary-data protocol, and how much deployed-model quality is lost through quota candidate generation. Chapter 3 translates those questions into controlled experiments.

剩下的经验问题是, 在一个二次数据协议下, 一个小型的结结结SigLIP特性关系头是否能够平衡歧视, 完成排名, 运行时间和审计能力, 以及通过配额候选生成部署模型质量有多大损失。(原文数值: 3。)

2.11 总结

Cove has been positioned as a content-based compatibility and constrained-search system rather than a conventional personalised collaborative recommender. The literature shows a progression from pairwise visual relations to sequence, type-aware, contextual and attention-based modelling. It supports frozen vision-language features as a resource-conscious experimental layer while requiring explicit treatment of transfer, labels, evaluation and bias.

科夫已经被定位为基于内容的兼容性和有限搜索系统而不是传统的个性化协作推器。灯光显示了从双向视觉关系到序列、类型感知、文本和关注的建模的进步。它支持冻结的视觉-语言的表现,作为一个资源意识的实验层,同时需要明确处理转移、标签、评估和偏见。

English

Compatibility classification, completion ranking, calibration, search completeness and user benefit are distinct claims. The first four can be examined with secondary data and system experiments; user benefit cannot be claimed without participant evidence. Chapter 3 specifies the methodology derived from these boundaries.

中文

兼容性分类, 完成排名, 校准, 搜索完整性和用户益处是不同的索赔。前四个可以通过二次数据和系统实验来检查; 没有参与者证据, 用户益处不能索赔。第三章具体说明了这些界限所衍生的方法。(原文数值: 3。)

Chapter 3: Methodology

第三章 方法

3.1 Research approach

3.1 研究方法

English

This project used a quantitative secondary-data approach combined with software-system evaluation. The investigation examined whether a lightweight model could learn outfit compatibility from published fashion data and whether its scores could support a practical recommendation application. The research therefore covered both model performance and the behaviour of the complete recommendation process.

中文

该项目采用了量化二次数据方法, 结合软件系统评估。调查调查调查是否可以从出版的时尚数据中学习轻量级模型的穿搭兼容性, 以及是否其分数可以支持实践推应用。因此, 研究涵盖了模型性能和整个推过程的行为。

English

The project did not conduct interviews, questionnaires or other participant research. It consequently does not claim that the recommendations improve user satisfaction or represent universally accepted fashion choices. The available evidence is limited to offline model evaluation, controlled comparisons and functional system testing.

中文

该项目没有进行访谈, 问卷或其他参与者研究。因此, 该项目并没有声称推荐改善用户满意度或代表普遍接受的时尚选择。可用的证据仅限于离线模型评估, 控制比较和功能系统测试。

English

The complete technical route is shown in Figure 3.1. It separates secondary-data preparation and frozen feature extraction from compatibility learning, controlled evaluation and application deployment.

中文

完整的技术路线在图3.1中显示。它将二次数据准备和冻结期功能提取与兼容性学习, 控制评估和应用部署分开。

English

Polyvore disjoint Four-position mapping Frozen SigLIP Processed embeddings images and metadata and negative construction feature extraction and outfit tensors Compatibility-head Validation selection Selected checkpoint training and calibration and abstract prototypes Baselines, FITB, ablation, Constraint filtering and Cove recommendation bias and generalisation tests exact candidate search prototype Figure 3.1: Technical route from secondary fashion data to the evaluated prototype

中文

多体分离四位映射 结冰的SigLIP 处理嵌入式图像和元数据以及负面的建筑特征提取和装饰子 兼容头验证选择 选择检查点训练和校准和抽象原型 基线, FITB, 消融, 限制过和布推偏见和通用测试 准确的候选搜索原型图3.1图: 从二级时尚数据到评估原型的技术路径

3.2 Data and ethical scope

3.2 数据和道德范围

English

The main model was trained and evaluated using the published Polyvore Outfits dis-joint dataset described by Vasileva et al. (2018). The disjoint version was selected because the same product does not appear across its main training, validation and test divisions. This provides a more demanding evaluation of the model's ability to work with previously unseen garments.

中文

主要模型是通过Vasileva et al. (2018) 描述的发表的Polyvore Outfits分组数据集进行训练和评估的。由于该产品在其主要训练、验证和测试部门中不出现，因此选择了分组版本。这提供了更具要求的评估模型与以前未见过的服装工作的能力。

English

The application supports four clothing positions: an inner top, an optional outer layer, a bottom and shoes. After unsupported or incomplete examples were removed, the processed data contained 18,243 training, 3,274 validation and 16,062 test examples. Positive examples came from outfits retained in the published dataset, while negative examples were constructed by replacing a garment with another item from the same broad category.

中文

该应用程序支持四个服装位置：内顶，外层，底层和鞋子。在删除未支持或不完整的例子后，处理数据包含了18243个训练，3274个验证和16062个测试样本。积极的例子来自于保留在公布的数据集中的服装，而负面的例子是通过更换同一类型的其他物品来构建的。（原文数值：16,062、18,243、3,274。）

English

Polyvore was used to learn compatibility relationships. The separately obtained Zara dataset was not added to these compatibility labels. It was used only to examine garment recognition and differences between data sources. User-uploaded photographs remain on the local device and were not used as research data.

中文

聚体用于学习兼容性关系。单独获得的Zara数据集没有被添加到这些兼容性标签中。它仅用于检查服装识别和数据来源之间的差异。用户上传的照片仍然存在于本地设备上，并没有被用于研究数据。

3.3 Recommendation-model development

3.3 推模式的发展

English

SigLIP was used to convert each garment image into a numerical visual representation (Zhai et al., 2023). SigLIP itself was not retrained. Its existing parameters were frozen, and its image representations were calculated before the compatibility model was trained. This substantially reduced the amount of computation required and allowed every comparison model to use the same visual information.

中文

利普被用来将每一个服装图像转换为数字视觉表示（Zhai et al., 2023）。SigLIP本身没有重新训练。其现有参数被冻结，其图像表示在兼容性模型训练之前计算。这显著减少了所需的计算量，并允许每个比较模型使用相同的视觉信息。

English

A small neural model was then trained to assign one compatibility score to a complete outfit. The model combines two forms of information: an overall summary of the garments and the visual differences between garment pairs. Clothing-position information is also supplied so that, for example, the relationship between a top and shoes can be distinguished from the relationship between two other positions.

中文

随后, 一个小的神经模型被训练为一个完整的服装分配一个兼容性分数. 该模型结合了两种信息形式: 服装的总结和服装对之间的视觉差异.

English

The compatibility head contains approximately 379,000 trainable parameters. It was trained to distinguish positive outfits from constructed negative outfits. Although its output is later used to rank recommendations, the model is trained as a compatibility classifier rather than as a direct user-preference predictor.

中文

兼容性头包含约379,000个可训练的参数. 它被训练以区分积极的服装与构建的负面服装. 尽管其输出后来用于排名推荐, 但该模型被训练为兼容性分类器而不是直接的用户偏好预测器.

English

Figure 3.2 shows the implemented compatibility head. The trainable component receives frozen item embeddings and a position mask, then combines a masked outfit summary with explicit pairwise differences before producing one compatibility logit.

中文

图 3.2 显示了实现的兼容性头. 可训练的组件重新接收了结的项目嵌入和位置面具, 然后在产生一个兼容性逻辑之前将一个隐藏的服装总结与明确的双向差异结合起来.

English

Training was limited to 12 epochs and used validation performance to select the saved checkpoint. Three predefined random seeds were used to examine whether results depended heavily on one training run. The test set was kept separate from model selection.

中文

训练仅限于12个时代, 用于选择保存的检查点使用验证性能. 使用三个预定义的随机种子来检查结果是否严重依赖于一个训练运行. 测试组与模型选择保持分开.

English

The processed labels were already balanced, so the final model did not apply additional Four 768-dimensional Layer normalization, learned embeddings linear projection position embedding and GELU beddings Six absolute position mask pairwise Masked mean difference outfit summary vectors and masked mean Concatenate Compatibility Two-layer MLP global and logit with GELU pairwise and calibration and dropout summaries rated score Figure 3.2: Architecture of the task-specific lightweight compatibility model positive-class weighting. Alternative weighting by outfit length was tested but did not satisfy the complete acceptance criteria. The unweighted model was therefore retained. Detailed settings and the rejected weighting experiment are reserved for the appendices.

中文

已有平衡的标签, 在最终模型下已有了附加的四768维层-合标志物质调整, 学习嵌入式线性投影位置-加位-和GELU床位Six absolute 面具对面 面具面具面具面具差异 面具面具面具平均值 面具对面具兼容性 双层MLP全球和GELU对面和卡比和降落总结评分分数图3.2: 任务特定轻量兼容性模型的架构 面具尺寸的替代权重测试, 但未满足完全的接受标准. 因此, 保留了未加权的模型. 详细设置和拒绝的权重实验为附件保留.

3.4 Comparative experiments

3.4 较量实验

English

The proposed model was first compared with random ranking and a non-learned SigLIP similarity baseline. These controls test whether the model performs above chance and whether learning garment relationships adds value beyond the frozen visual representation.

中文

推荐的模型首先与非学习的SigLIP相似性基线进行了比较。这些控制测试模型是否超出机会的性能，以及学习服装关系是否超出了冻结的视觉表示的增值。

English

Three further models represented established directions in outfit-recommendation re-search: a bidirectional LSTM based on Han et al. (2017), a type-aware pairwise model based on Vasileva et al. (2018), and a Transformer-based set model informed by Sarkar et al. (2023). They used the same processed data, frozen image representations and training conditions as the proposed model. They are controlled implementations of the main architectural ideas rather than exact reproductions of the published systems.

中文

另外三个模型代表了穿搭推荐重新搜索的既定方向：基于汉等人（2017）的双向LSTM，基于瓦西莱娃等人（2018）的类型感知双向模型，以及基于萨尔卡等人（2023）的变压器集模型。它们使用了与拟议模型相同的处理数据，结的图像表示和培训条件。它们是主要建筑理念的控制实现而不是发表的系统的精确复制。

English

Two ablation experiments were also conducted. One removed clothing-position information, and the other removed the module that explicitly compares garment pairs. The purpose was to determine whether these components made a measurable contribution when the rest of the experimental conditions remained unchanged.

中文

另外，还进行了两次除实验。一个取消了服装位置信息，另一个取消了明确比较服装对的模块。目的是确定这些组件是否在其他实验条件保持不变时产生了可测量的贡献。

3.5 Model and system evaluation

3.5 模型和系统评估

English

Compatibility classification was evaluated using several complementary measures. ROC-AUC measured how reliably positive outfits received higher scores than negative outfits (Fawcett, 2006). Accuracy, balanced accuracy, precision, recall, F1 and the confusion matrix were used to examine classification behaviour at the selected threshold. Calibration measures were reported separately because a model can rank examples correctly while still producing overconfident scores (Guo et al., 2017).

中文

根据测量结果，可测量结果是如何对正面服装获得比负面服装更高的分数（Fawcett, 2006）。精度，平衡精度，精度，回忆，F1和混矩阵用于检查在选择门时的分类行为。校准措施分别报告，因为模型可以正确排名示例，同时仍然产生过度自信的分數（Guo et al., 2017）。

English

Recommendation performance was examined through the official fill-in-the-blank task. One garment is removed from an outfit and the model must select the correct replacement from four candidates. The correct answer was matched through the outfit identifier because the answers in the disjoint data are shuffled. Only questions compatible with the project's four clothing positions were retained, and the retained proportion is reported with the result.

中文

推荐的性能通过官方填写白色任务进行了测试。从一个服装中取出一个服装，模型必须从四名候选中选择正确的替换。正确的答案通过服装标识器匹配，因为分类数据中的答案是混的。仅保留了与项目相符的四个服装位置的问题，并且保留的比例与结果进行报告。

English

Confidence intervals were estimated by repeatedly resampling the evaluation questions (Efron and Tibshirani, 1993). The same questions and random seeds were used when comparing the complete model with an ablation, allowing their differences to be assessed directly. Training, validation and test behaviour were also compared to identify evidence of overfitting.

中文

通过重复测试问题进行测量，估计了信心间隔（Efron和Tibshirani, 1993）。在将完整模型与抽象进行比较时，使用相同的问题和随机种子，从而直接评估它们的差异。（原文引用：（Efron and Tibshirani, 1993）。）

English

The complete application was tested separately from the classifier. Candidate availability was checked under different weather and style conditions. Search experiments examined whether the previous quota-based method could miss higher-scoring combinations and measured the time required to search the complete filtered candidate space.

中文

根据该研究的结果，研究人员发现，在一个特定的时间内，一个特定的应用程序可以被测试，一个特定的应用程序可以被测试。

English

Figure 3.3 groups the evidence used to assess the model and the surrounding recommendation process. No single metric is treated as sufficient for deployment or for a claim about recommendation quality.

中文

图3. 3将用于评估模型和周围的推过程的证据组合在一起。没有一个指标被视为充分用于部署或对推质量的声明。

English

Disjoint train, validation Selected model and held-out test data controlled alternatives Classification Completion ranking Robustness System behaviour AUC and threshold metrics FITB, MRR and NDCG seeds, ablation, calibration coverage, search and runtime Evidence with confidence intervals, coverage and stated boundaries Model selection, deployment gate and prioritised limitations Figure 3.3: Evaluation framework connecting model, ranking and system evidence

中文

脱训练，验证 选择模式和数据测试控制方案 类别 完成 强度排名系统行为 AUC和门测量 FITB, MRR和NDCG种子，消融，校准覆盖，搜索和运行时间 证据与信心间隔，覆盖和确定的边界 模型选择，部署门和优先限制图3. 3：评估框架连接模型，排名和证据系统

3.6 Experimental controls and decision rules

3.6 实验控制和决策规则

English

The experiments were designed so that each comparison changed a defined part of the pipeline. Architecture comparisons used the same processed examples, frozen SigLIP features, epoch limit, optimiser settings and predefined random seeds. This does not make the models identical in their learning behaviour, but it reduces the risk that a reported difference is caused by one model receiving different visual inputs or a more favourable data split. Parameter counts and recorded training times were reported to place performance in the context of model size and implementation cost. Training time was not treated as a hardware-independent benchmark.

中文

实验设计以使每个比较改变了管道的定义部分。建筑比较使用相同的加工示例, 结的SigLIP功能, 时代限制, 优化设置和预定义的随机种子。这并不使模型在学习行为上相同, 但它降低了报告的差异是由一个模型接收不同的视觉输入或更有利的数据分区引起的风险。参数计数和记录的训练时间被报告以将性能放在模型大小和实施成本的背景下。训练时间并没有被视为硬件独立的基准。

English

Model selection was separated from final reporting. Validation AUC selected the saved checkpoint, and the test data were then used to estimate performance on unseen ex-amples.

The three-seed architecture experiment examined variation from initialisation and training order. Multiple runs cannot remove every source of benchmark uncertainty, but they provide more reliable evidence than one favourable run (Bouthillier et al., 2021). Paired bootstrap comparisons reused the same resampled test examples for the two models being compared. This preserves the relationship between their predictions and is more informative than treating the scores as unrelated observations (Dietterich, 1998; Efron and Tibshirani, 1993).

中文

验证ACC选择了保存的检查点, 然后测试数据被用来估计未见的示例的性能。三种种结构实验测试了从初始化和训练顺序的变化。多次运行不能消除每个基准不确定性的来源, 但它们提供比一个有利的运行更可靠的证据 (Bouthillier等人, 2021)。对合的启动杆比较再次使用相同的重新样本测试示例。这保留了它们的预测之间的关系, 并且比较处理分数为无关的观察 (Dietterich, 1998; Efron and Tibshirani, 1993)。(原文引用: (Bouthillier et al., 2021); (Dietterich, 1998; Efron and Tibshirani, 1993)。)

English

Decisions about model changes used more than overall AUC. The length-weighting experiment had to retain overall discrimination, avoid reducing macro-level performance and improve the weakest length group. It was rejected when the weakest group deteriorated, despite a small improvement in overall mean AUC. The ranking-loss experiment was also not adopted because it did not improve the retained completion result. These decision rules reduce the temptation to select a configuration from one favourable number after seeing the results. They remain specific to the tested alternatives and do not imply that all weighting or ranking losses are unsuitable.

中文

模型变化的决定使用了超过总体AUC。长度权重实验必须保持总体歧视, 避免减少宏观水平的性能, 改善最弱的长度组。它被拒绝了当最弱的组恶化, 尽管总体平均AUC的小幅改善。排名损失实验也没有被采用, 因为它没有改善保留的完成结果。这些决策规则减少了在看到结果后从一个有利数字中选择配置的诱惑。它们仍然是针对测试的替代品的具体, 并不意味着所有权重或排名损失都是不合适的。

English

The search comparison also kept the scoring objective and candidate artefact fixed. Exact enumeration and quota-limited retrieval received the same held-out inputs and feasible candidate states. The exact path supplied the reference top results for the bounded space. Top-one recovery, top-ten recall, score loss and runtime then de-scribed what the quota path gained and lost. Human preference was not included in this comparison, so a higher model score was interpreted only as better optimisation of the deployed compatibility objective.

中文

搜索比较还保持了得分目标和候选文物固定。精确的编号和配额限制检索得到了相同的持久输入和可行的候选状态。精确的路径为有限空间提供了参考顶级结果。前一回复, 前十回忆, 得分损失和运行时间然后删除了配额路径获得和损失的内容。人类偏好没有被纳入这次比较, 因此更高的模型得分仅被解释为更好的实现部署兼容性目标的优化。

3.7 Application integration, bias and reproducibility

3.7 应用集成, 偏见和可复制性

English

Therecommendationinterfacepresentsbroadclothingconceptssuchasablackcasual jacket or white trainers rather than claiming that the model has selected a particular commercial product. These concepts are derived from garments in the Polyvore train-ing split and are displayed as coloured icons. Photographs are used only for garments uploaded or saved by the user.

中文

推界面介绍了黑色休克或白色训练服等广泛的服装概念, 而不是声称模型选择了特定商业产品。这些概念来自 Polyvore训练区内的服装, 并显示为彩色图标。照片仅用于用户上传或保存的服装。

English

Weather, clothing position and the selected menswear or womenswear range are treated as explicit constraints. The application does not infer gender from a photo-graph. Because pure menswear training examples are limited, the same compatibility model is used for both ranges and no claim is made that their reliability is equal.

中文

由于纯男子服装训练示例有限, 因此对两种系列都使用相同的兼容性模型, 并没有说它们的可靠性是相同的。

English

Reproducibility records include random seeds, model settings, training histories, soft-ware versions, file hashes and evaluation outputs. Automated tests cover data shapes, missing clothing positions, corrected fill-in-the-blank labels, clothing-range filtering, recommendation determinism and search safety limits. These records allow the re-ported experiments to be checked without relying only on the running user interface.

中文

复制性记录包括随机种子, 模型设置, 训练历史, 软件版本, 文件哈希和评估输出。自动测试涵盖数据形状, 缺失服装位置, 纠正填空标签, 服装范围过, 推确定性和搜索安全限制。这些记录允许重新报告的实验进行检查, 而不仅仅依赖运行用户界面。

English

The software was organised so that training, evaluation and application inference could be run independently. Processed tensors and frozen embeddings were stored as explicit artefacts, rather than being recreated silently during each training run. Model checkpoints include the configuration required to reconstruct the compatibility head, while evaluation scripts write structured result files used by the tables in Chapter 4. This separation reduces the risk that interface state or later UI changes alter a reported experiment.

中文

软件的组织是这样的, 培训, 评估和应用推断可以独立运行. 处理的子结和嵌入式被存储为明确的文物, 而不是在每个培训运行中默默地重建. 模型检查点包括重建兼容性头所需的配置, 而评估脚本则会写出由第4章表所使用的结构化结果文件. 这种分离减少了接口状态或后续UI变化改变报告的实验的风险.

English

Integration used a guarded loading path. The application first checks whether the candidate artefact, model checkpoint and required metadata are present and compatible. If they are available, the learned score participates in ranking; otherwise the existing rule procedure is used and the fallback reason is recorded. The model was not allowed to replace category, weather or clothing-range validation. This design made it possible to test learned recommendation without making the whole application dependent on one model file. It also distinguishes reproducibility of an experimental checkpoint from availability of a user-facing feature.

中文

集成使用了一个保护的加载路径. 应用程序首先检查可查的文物, 模型检查点和所需的元数据是否存在和兼容. 如果它们可用, 学习得分参与排名; 否则使用现有的规则程序并记录反弹原因. 该模型不允许取代类别, 天气或服装范围验证. 这种设计使得测试学习推荐能够在不使整个应用程序依赖于一个模型文件的情况下进行. 它还区分了实验检查点的可再生性与用户面向的功能的可用性.

English

The runtime sequence is shown in Figure 3.4. The selected garment remains fixed, contextual constraints define the feasible candidate space, and the learned model ranks complete combinations. A deterministic content-based path is retained only as an operational fallback when the required model artefacts cannot be loaded.

中文

运行时间序列在图3.4中显示. 选择的服装仍然是固定的, 文本限制定义了可行的候选空间, 学习模型排列了完整的组合. 基于内容的确定性路径只保留在无法加载所需的模型文物时作为操作倒退.

English

Selected Type, col- Fix selected Model garment our, style yes item; filter by artefacts or local ward- and SigLIP weather, style available? robe piece embedding and range no Exact enu- Deterministic meration and Primary out-content batched compat-fit, diverse ranker fallback ibility scoring alternative and replacements Figure 3.4: Runtime recommendation and fallback flow Table 3.1 records the responsibility of each core application module. This division prevents the compatibility score from being described as if it also performs recognition, weather validation or storage.

中文

选择类型, col- 修复选择的模型服装, 样式是的项目; 按原件或当地区和siglip天气进行过, 样式可用? 服装件嵌入和范围无精确的精确的确定性和主要外观内容组合合适, 多元化的排列器倒退能力评分替代品和替代品图3.4: 运行时间推荐和倒退流量图3.1记录每个核心应用模块的责任. 这一分类防止对描述的设备进行认可, 气象验证或存储.

English

Table 3.1: Core functions of the Cove prototype Module Main input and out- Responsibility put Garment analysis Selected crop → type, Interprets the item supplied by the user and pre-colour, style and em- pares its model representation. bedding Candidate control Context and fixed item Applies position, weather, style and clothing- → feasible states range constraints. Compatibility Completecandidates→ Scores relational coherence using the trained ranking ordered scores lightweight head. Wardrobe man- Piece and outfit edits Saves, reuses, renames, groups and removes age- ment → local records and im- local wardrobe content. ages Fallback Recognised item and Preserves basic operation when learned arte- context → rule-ranked facts are absent or incompatible. outfit The prototype does not use a relational database. Its persistent state is held in JSON documents and local image files, while model and research outputs are separate arte- facts. Figure 3.5 presents these logical relationships without implying tables, foreign keys or a database engine that are not present in the implementation.

中文

表3.1: 体原型模块核心功能 主要输入和外出 - 责任置 服装分析 选择的裁剪图 →类型, 解释用户提供的物品和预颜色, 风格和模特表现. 床单 候选控制 背景和固定物品 适用于位置, 天气, 风格和服装 →可行状态范围限制. 兼容性 完整候选→ 通过训练有素的排名来得分关系一致性 排序分数轻量头. 衣柜人- 片和服装编辑保存, 重用, 更名, 组组和删除 年代 → 地方记录和 im- 地方衣柜内容. 管理 倒退 识别项目和 保存基本操作当学习的模型产物 → 规则排名的事实是缺失或不兼容的. 服装 原型不使用关系数据库. 其持久状态是保持在 JSON 文件和本地图像文件, 而模型和研究输出是单独的文本. 图3.5 呈现这些逻辑关系, 不含表, 外键或数据库引擎, 不存在于实现.

English

Supplementary software-design views are provided in Appendix D. They include the implemented use cases, overall component structure, recommendation-module de- Modelartefacts wardrobe.json compatibilitycheckpointand version,pieces[],outfits[], abstractprototypecatalogue outfit_groups[] contains contains loadedby Piece Outfit Applicationruntime 0-4embeddedpiecevalues id,position,type,colour, id,title,mood,style,group, loadslocaluserstateand image_path,source,created_at fourpositionvalues,created_at availablemodelartefacts references readby wardrobe_images/ Researchoutputs user_preferences.json localPNGfilesreferenced evaluationJSON,manifests, documentedby uptosixsavedcities byPiece.image_path hashesandtraininghistories Figure 3.5: Logical relationships among local user data and model artefacts; the prototype uses files rather than a relational database Table 3.2: Local persistence and research artefacts File or artefact Principal contents Role and relationship wardrobe.json Pieces, outfits and outfit Local user store; each outfit contains groups valuesforthefoursupportedpositions. wardrobe_images/ User garment PNG files Image paths are referenced by saved piece records. user_preferences.json Saved cities Read by the application to restore local location choices. Compatibility checkpoint Model configuration, Reconstructstheselectedcompatibility state and metadata scorer at runtime. Abstract prototype arte- Candidate states and Supplies the product-independent can-fact frozen embeddings didate space for constrained search. Evaluation JSON and Metrics, histories, hashes Preserve the evidence chain; they are manifests and settings not user-profile records.

中文

附录D提供了补充软件设计的视图. 包含实施的使用情况, 整体组件结构, 推荐模块解调模块衣柜. json兼容性检查点和版本, 零件[], 外套[], 抽象原型目录外套_组[] 包含包含加载的零件外套应用程序运行时间0-4内置的零件值ID, 位置, 类型, 颜色, ID, 标题, 情绪, 风格, 组, 载荷本地用户状态和图像_路径, 来源, 创建_在四个位置值, 创建_在可用性时_模块外观引用阅读衣柜_图像/图像/图像_偏好. jPNG文件评测JSON, 引用, 记录, 由用户数据中心进行测试. n模型文物; 原型使用文件而不是关系数据库表 3.2: 本地持久性和研究文物 文件或文物 主要内容 角色和关系 衣柜. json 片, 服装和服装 本地用户商店; 每个服装包含四个支持的位置的组值. 衣柜_图片/用户服装 PNG 文件 图像路由保存的件记录引用. 用户_偏好. json 保存的城市 通过应用程序来恢复本地位置选择. 兼容性检查点 模型配置, 重建选择的兼容性状态和运行时的元数据得分器. 抽象原型艺术- 候选状态和供应产品独立的结嵌入式数据空间限制搜索. 评估 JSON 和测量, 历史, 哈希保存证据链; 它们是表现和设置, 而不是用户配置文件记录. (原文数值: 3.5.)

English

dependencies, a logical entity-relationship view and traceable user stories. The diagrams describe the current prototype rather than an assumed enterprise architecture.

中文

它们描述了当前的原型而不是假设的企业架构。

English

Result artefacts and checkpoints are larger than ordinary source files and may not always be suitable for repository distribution. File hashes and documented generation commands therefore provide an additional check that a locally held artefact corresponds to the evaluated version. This approach does not replace archival storage or full environment preservation, but it provides a practical chain from processed data to training history, evaluation output and deployed configuration.

中文

结果文物和检查点比普通源文件大，并可能不适合存储库分布。因此文件哈希和记录生成命令提供了额外的检查，即本地持有的文物与评估版本相匹配。这种方法不取代存档存储或完整的环境保护，但它提供了从处理数据到培训历史，评估输出和部署配置的实用链。

3.8 Research risks and mitigation

3.8 研究风险和减轻

English

The investigation involved data, methodological, ethical and operational risks. These were considered during development rather than being listed only after the experiments. The first concerned data provenance and permitted use. The project therefore used documented secondary datasets and kept their roles separate. Polyvore supplied compatibility labels, whereas Zara was limited to recognition and feature-domain diagnostics. User photographs remained local and were not incorporated into the research dataset. This boundary reduces privacy exposure, but it does not establish equal provenance or representation quality for every image. Any later public release would require a separate check before third-party images or large derived artefacts could be redistributed.

中文

调查涉及数据、方法、伦理和运营风险。这些风险在开发过程中被考虑，而不是仅仅在实验后列出。第一个涉及数据来源和允许使用。因此，该项目使用了记录的二次数据集，并保持了它们的角色分开。多体提供了兼容性标签，而Zara仅限于识别和特征域诊断。用户照片仍然是本地，并没有被纳入研究数据集。这一边界减少了隐私暴露，但它并没有为每个图像建立相同的来源或表示质量。任何后期公开发布都需要在第三方图像或大型衍生品重新分配之前进行单独检查。

English

A second risk was data leakage. Product overlap between training and test data could allow a model to recognise garments seen during training instead of learning relationships that transfer to unseen items. The published disjoint split was used to reduce this risk, and candidate concepts were derived from the training division rather than the test division. Model selection and temperature scaling used validation data, while the test set was reserved for final evaluation. Repeatedly changing the model after inspecting test results could still weaken test independence. The current test results are therefore treated as final evidence for the selected model rather than as a further tuning target.

中文

另一种风险是数据泄露。培训和测试数据之间的产品重叠可以使模型能够识别在培训期间看到的服装，而不是转移到未见物品的学习关系。发表的分离分离被用来减少这种风险，并且候选概念来自培训部门而不是测试部门。模型选择和温度扩展使用验证数据，而测试集则保留用于最终评估。检测结果后重复改变模型仍然可能会削弱测试独立性。因此，当前的测试结果被视为选择模型的最终证据，而不是进一步调整目标。

English

The labels create a further interpretive risk. Positive examples are observed Polyvore outfits, whereas negatives are constructed by replacing an item within a broad category. A constructed negative is not proof that an outfit is inherently unacceptable; it is a reproducible device for learning distinctions within this dataset. Similarly, official FITB evaluation was restricted to questions represented by the four supported positions and available embeddings. Coverage is reported with accuracy so that performance on the retained subset is not presented as a complete benchmark result. This follows the principle that recommender metrics must be interpreted in relation to the task and sampling protocol (Herlocker et al., 2004).

中文

标签创造了进一步的解释风险。积极的例子是观察到的多种服装,而负面的构建是通过更换一个广泛的猫格里内的项目。构建的负面不是证明服装本质上是不可接受的证据;它是一种可复制的工具来学习在这个数据集中的区别。同样,官方的FITB评估被限制在由四个支持的位置和可用的嵌入式表示的问题。覆盖率是准确报告的,以便保留的子集上的性能不被呈现为完整的基准结果。这遵循推荐指标的原则,必须与任务和采样协议相关地解释 (Herlocker等人, 2004)。(原文引用: (Herlocker et al., 2004)。)

English

Overfitting and unstable model selection were addressed by recording training, validation and test behaviour separately, using three random seeds for the controlled architecture comparison and selecting checkpoints from validation AUC. Confidence intervals were estimated for the principal test and paired comparisons. Early stopping limits continued fitting after validation performance stops improving, but does not remove a generalisation gap (Prechelt, 1998). The results therefore report both the close validation-test result and the larger train-test difference.

中文

通过单独记录训练,验证和测试行为,使用三个随机种子进行控制架构比较,并从验证AUC中选择检查点来解决过度安装和不稳定的模型选择。主要测试和对比测试的信心间隔被估计。验证性能停止改善后,早期停止制限制继续安装,但并没有消除泛化差距 (Prechelt, 1998)。因此,结果报告了近距离验证测试结果和火车测试差异。

English

Representation across clothing ranges was unequal. The retained FITB questions and pure training examples were concentrated in womenswear, so the small menswear subgroup could not support a reliable fairness comparison. The interface asks users to choose a clothing range and does not infer gender from an image. Menswear evidence is reported as sparse coverage rather than proof of equal reliability. This does not resolve representational bias, but it avoids automated gender classification and makes the evidential boundary explicit (Keyes, 2018; Mehrabi et al., 2021).

中文

服装范围内的代表性不平等。保留的FITB问题和纯训练示例集中在女性服装上,因此小男装子组无法支持可靠的公平性比较。接口要求用户选择服装范围,而不根据图像推断性别。男装证据被报告为稀有覆盖,而不是平等可靠性的证据。这不会解决代表性偏见,但它避免了自动性分类,并使证据界限明确 (Keyes, 2018; Mehrabi et al., 2021)。

English

Finally, the deployed process could fail if model files, embeddings or candidate records were unavailable, or if constraints removed every candidate for a position. Release checks, deterministic seeds and recorded artefact paths support reproducibility. Slot, weather and clothing-range constraints reject infeasible combinations, while a deterministic rule-based fallback allows the application to continue when learned artefacts cannot be loaded. The fallback improves operational resilience; it is not evidence that its recommendations are stylistically correct. These mitigations make failure conditions more visible and bounded, without implying that technical testing removes all social or external-validity risks.

中文

最后, 如果模型文件, 嵌入式或候选记录不存在, 或者如果限制删除了每个候选. 释放检查, 确定性种子和记录的文物路径支持可复制性. 位置, 天气和服装范围的限制拒绝不可行组合, 而基于确定性主义规则的倒退允许应用程序继续运行, 当未能加载学习的文物时. 倒退提高了运营弹性; 这不是证明其推荐是风格正确的证据. 这些减轻使故障条件更加明显和有限, 而无需暗示技术测试消除了所有社会或外部有效性风险.

3.9 Summary

3.9 总结

English

The methodology combines a frozen visual encoder, a lightweight compatibility model and a constrained recommendation process. Controlled baselines, repeated training runs, fill-in-the-blank evaluation, ablation experiments and system tests were used to examine different aspects of performance. The following chapter reports the corresponding results without extending them to unsupported claims about universal fashion quality or user satisfaction.

中文

该方法结合了冻结的视觉编码器, 轻量级兼容性模型和有限的推过程. 用于检查性能的不同方面, 使用了控制的基线, 重复训练运行, 填空评估, 除实验和系统测试. 下一章报告了相应的结果, 但没有扩展到关于普遍时尚质量或用户满意度的无证据指控.

Chapter 4: Results

第四章 结果

4.1 Introduction

4.1 引言

English

This chapter reports the compatibility, architecture, ablation, overfitting, bias and candidate-search results. Results are separated according to their experimental protocol. In particular, the selected compatibility model, the three-seed architecture comparison and the candidate-search experiment are reported independently because they answer different questions.

中文

本章报告了兼容性、架构、消融性、过拟合性、偏见性和候选搜索结果。结果根据其实验性结果分开。特别是，选择的兼容性模型、三种建筑比较和候选搜索实验独立报告，因为它们回答不同的问题。

4.2 Selected compatibility-model performance

4.2 选择的兼容性模型性能

English

The selected compatibility model was evaluated on 16,062 held-out test examples containing equal numbers of positive and negative rows. It achieved a test ROC-AUC of 0.8583, with a 95% confidence interval from 0.8524 to 0.8641. Accuracy and balanced accuracy were both 0.7664. Table 4.1 reports the remaining classification results.

中文

选择的兼容性模型在 16,062 个保持测试样本上进行了评估，其中有相同数量的正和负行。它达到 0.8583 的测试 ROC-AUC，95% 的可靠性间隔从 0.8524 到 0.8641。准确性和平衡准确性均为 0.7664。表 4.1 报告了剩余的分类结果。

English

Table 4.1: Held-out performance of the selected compatibility model
Metric Result ROC-AUC 0.8583 95% AUC confidence interval 0.8524-0.8641 Average precision 0.8558 Accuracy 0.7664 Balanced accuracy 0.7664 Precision 0.7240 Recall 0.8610 F1 0.7866 Specificity 0.6718 Matthews correlation coefficient 0.5426 Brier score 0.1599 At the 0.5 threshold, the model correctly classified 5,395 negative and 6,915 positive outfits. It classified 2,636 negative outfits as positive and 1,116 positive outfits as negative. Recall was therefore higher than specificity at this threshold.

中文

表 4.1：选择的兼容性模型的保持性能
测量结果 ROC-AUC 0.8583 95% AUC 信心间隔 0.8524-0.8641 平均精度 0.8558 精度 0.7664 均衡精度 0.7664 精度 0.7240 提醒 0.8610 F1 0.7866 具体性 0.6718 马修斯相关系数 0.5426 Brier 评分 0.1599 在 0.5 门上，该模型正确分类了 5,395 负面服装和 6,915 积极服装。它分类了 2,636 负面服装为积极服装和 1,116 积极服装为负面服装。因此，提醒在这个门上比特征高。

English

A separate fixed-seed baseline experiment produced an AUC of 0.4940 for deterministic random scores and 0.7012 for mean pairwise SigLIP cosine similarity. The learned head in that experiment reached 0.8573. These values came from an earlier controlled baseline run and are reported separately from the selected model result of 0.8583.

中文

单独的固定种子基线实验产生了0.4940的AUC, 用于确定性随机分数和0.7012的平均对称SigLIP共数相似性。该实验的学习头达到0.8573。这些值来自早期的控制基线运行, 并且与选定的模型结果分别报告0.8583。

4.3 Controlled architecture comparison

4.3 控制式建筑比较

English

Each learned architecture was trained with seeds 42, 7 and 123. The proposed model achieved the highest mean test AUC and the smallest AUC standard deviation across the three runs (Table 4.2).

中文

每个学习的架构都采用了42, 7和123种子进行训练。拟议的模型在三个运行中实现了最高的平均测试AUC和最小的AUC标准偏差 (表4.2)。

English

Table 4.2: Controlled three-seed architecture comparison

Model	Parameters	Time (s)	Mean AUC	SD
Bidirectional LSTM	397,185	96.87	0.7928	0.0060
Type-aware pairwise	591,489	7.99	0.7918	0.0129
Set Transformer	374,145	7.03	0.7744	0.0133
Proposed model	379,265	4.81	0.8596	0.0009

The proposed model's mean balanced accuracy was 0.7742 and its mean Matthews correlation coefficient was 0.5510. Mean balanced accuracy was 0.7138 for the bidirectional LSTM, 0.7176 for the type-aware model and 0.7034 for the set Transformer.

中文

控制式三种结构比较模型参数时间 (s) 中平均AUC SD双向LSTM 397,185 96.87 0.7928 0.0060 型号意识双向 591,489 7.99 0.7918 0.0129 设置变压器 374,145 7.03 0.7744 0.0133 拟议模型 379,265 4.81 0.8596 0.0009 拟议模型平均平衡精度为0.7742 及其平均马修斯相关系数为0.5510。(原文数值: 0.7034、0.7138、0.7176、4.2。)

English

Paired bootstrap comparison of the three-seed mean predictions gave an AUC difference of 0.0700 between the proposed model and the bidirectional LSTM (95% interval 0.0650-0.0753). The difference from the type-aware model was 0.0634 (0.0578-0.0694), while the difference from the set Transformer was 0.0801 (0.0750-0.0858). These results compare controlled implementations on the project's processed data rather than the headline results in the original publications.

中文

两种种子平均预测的配对启动线比较给出了拟议模型和双向LSTM之间的AUC差异0.0700 (95%间隔0.0650-0.0753)。与型号意识模型的差异为0.0634 (0.0578-0.0694), 而与集型变压器的差异为0.0801 (0.0750-0.0858)。这些结果比较了项目处理数据的控制实现而不是原始出版物标题结果。

4.4 Fill-in-the-blank and ablation results

4.4 填空和除结果

English

Of 15,145 official disjoint FITB questions, 4,468 were compatible with the project's four-position representation, giving coverage of 29.50%. Other questions were excluded because of unsupported answer positions, insufficient supported garments, duplicate positions or an answer position already occupied in the question.

中文

在15145个官方FITB问题中, 4468个符合项目4个位置的代表性, 占29.50%。其他问题因未支持的答案位置, 支持服装不足, 复制位置或已经在问题中占据的答案位置而被排除在外。(原文数值: 15, 145、4, 468。)

English

The selected model achieved FITB accuracy of 0.6025. Its 95% bootstrap confidence interval was 0.5873-0.6175, compared with chance accuracy of 0.25. Mean reciprocal rank was 0.7705 and normalised discounted cumulative gain was 0.8289.

中文

选择的模型达到0.6025的FITB精度。其95%的启动带可靠性间隔为0.5873-0.6175, 而随机精度为0.25。平均交互级别为0.7705, 正常折扣累计收益为0.8289。

English

The structural ablation used the same 4,468 questions and three random seeds. Table 4.3 reports the mean accuracy and the paired difference between the complete model and each reduced variant.

中文

结构式抽象使用了相同的4,468个问题和三个随机种子。(原文数值: 4.3。)

English

Table 4.3: Structural ablation on the retained official FITB subset Configuration Mean accuracy Full minus variant Paired 95% interval Complete model 0.5994 - - Without 0.5947 0.0047 -0.0013-0.0106 clothing-position embedding Without pairwise 0.4966 0.1028 0.0896-0.1157 module The retained FITB data contained only 46 menswear questions, compared with 4,422 womenswearquestions. Menswearperformancewasthereforerecordedasacoverage limitation rather than treated as a reliable comparison between clothing ranges. The descriptive subgroup values are provided in the appendices.

中文

图4.3:保留的官方FITB子集结构式缩 配置中度准确性 完全减变量 合并 95%间隔 完整模型 0.5994 - - 没有 0.5947 0.0047 □0.0013-0.0106服装位置嵌入 没有对称 0.4966 0.1028 0.0896-0.1157模块 保留的FITB数据仅包含46个男装问题, 而女性服装问题则为4,422。男装性能在此之前记录了覆盖范围限制, 而不是作为服装范围之间的可靠比较。描述子组值在附件中提供。

4.5 Calibration and overfitting audit

4.5 校准和过拟合审计

English

The selected checkpoint reached its highest validation AUC at epoch 6. Training, validation and test AUC were 0.9699, 0.8598 and 0.8583 respectively (Table 4.4).

中文

选择的检查点在6期达到最高验证AUC。训练、验证和测试AUC分别为0.9699, 0.8598和0.8583 (表4.4)。

English

Table 4.4: Performance of the selected checkpoint by data split Split Examples AUC Accuracy F1 Training 18,243 0.9699 0.8997 0.9051 Validation 3,274 0.8598 0.7688 0.7845 Test 16,062 0.8583 0.7664 0.7866 The train-test AUC difference was 0.1116, while the absolute validation-test difference was 0.00145. Validation AUC declined by 0.0100 between the saved epoch and the last observed epoch.

中文

表4.4: 通过数据分隔选择的检查点的性能 分类 实例 AUC精度 F1 训练 18,243 0.9699 0.8997 0.9051 验证 3,274 0.8598 0.7688 0.7845 测试 16,062 0.8583 0.7664 0.7866 列车测试AUC差异为0.1116, 而绝对验证-测试差异为0.00145. 验证AUC在保存的时代和最后观察到的时代之间下降了0.0100.

English

Validation-only temperature scaling produced a temperature of 1.5760. Validation negative log-likelihood decreased from 0.5081 to 0.4776, while expected calibration error decreased from 0.0772 to 0.0588. Candidate ordering was unchanged by the calibration step.

中文

仅验证温度扩展产生了1.5760. 验证负日志概率从0.5081降低到0.4776, 而预期校准错误从0.0772降低到0.0588.

4.6 Outfit-length imbalance and feature-domain results

4.6 服装长度失衡和特征领域结果

English

The class labels were balanced. A separate experiment examined the imbalance between three-item and four-item outfits. Table 4.5 reports the three configurations across the same three seeds.

中文

类标签是平衡的. 一个独立的实验检查了三项和四项服装之间的不平衡. 4.5表报告了三种种子中的三个配置.

English

Table 4.5: Outfit-length weighting experiment Configuration Overall AUC Weakest-length AUC Group gap Unweighted 0.8596 0.8507 0.0393 Moderate length weighting 0.8601 0.8500 0.0433 Stronger length weighting 0.8553 0.8438 0.0485 Moderate weighting achieved the highest overall mean AUC but reduced the mean result for the weakest length group and increased the group gap. It therefore failed the predefined deployment gate, and the unweighted configuration was retained.

中文

表4.5: 服装长度权重实验 配置总体AUC 最弱长度AUC组差距 未权重0.8596 0.8507 0.0393 适度长度权重0.8601 0.8500 0.0433 强度长度权重0.8553 0.8438 0.0485 适度权重实现了最高的总体平均AUC, 但减少了最弱长度组的平均结果并增加了组差距. 因此它未达到预定义的部署门, 并且保持了未权重配置.

English

The feature-domain diagnostic compared 26,124 Polyvore test items with 300 Zara catalogue items. The domain classifier achieved cross-validated AUC of 1.000. Centroid cosine similarity was 0.7452, and the mean nearest-Polyvore similarity for the Zara items was 0.6528. The implications of this separability are considered in Chapter 5.

中文

区别诊断对26124个多体体试验项进行了对比, 其中300个是Zara目录项. 域分类器实现了1000个的交叉验证AUC. 中心体共体相似性为0.7452, Zara项的平均近距离多体体相似性为0.6528. (原文数值: 1.000、26,124.)

4.7 Candidate-search results

4.7 候选搜索结果

English

The product-independent candidate artefact contained 1,670 clothing concepts de-rived from the Polyvore training split, covering 23 garment types, 12 colours and nine styles. The selected womenswear range contained 1,508 concepts and the menswear range contained 162. Both ranges contained candidates for all four supported clothing positions.

中文

产品独立的候选艺术品包含了1670款脱离Polyvore训练分类的服装概念, 涵盖了23种服装类型, 12种颜色和9种风格。(原文数值: 1, 508、1, 670、162。)

English

The earlier retailer-derived candidate space contained 300 product records and 122 visible type-colour combinations. The product-independent representation contained 271 such combinations. Absolute compatibility scores across these candidate sources were not treated as directly comparable because their data distributions differ.

中文

早期的零售商衍生的候选空间包含300个产品记录和122个可见的字体颜色组合。产品独立的表示包含271个这样的组合。这些候选来源的绝对兼容性分数并没有被视为直接可比, 因为它们的数据分布不同。

English

Exact and quota-limited search were compared on 18 held-out inputs using the same product-independent candidate space and compatibility model. The quota-limited path recovered the exact top result in 27.78% of queries and had mean recall of 24.44% for the exact top ten. It examined a mean 3.41% of the candidate combina-tions.

中文

根据相同的产品独立的候选空间和兼容性模型, 对18个持久的输入进行了准确和配额限制搜索的比较。在27.78%的查询中, 配额限制的路径获得了准确的最高结果, 对准确的前十个中平均回调为24.44%。(原文数值: 3.41%。)

English

The mean score loss relative to the highest score found by exact search was 0.0179, with a 95% bootstrap interval of 0.0063-0.0357. Thirteen of the 18 queries had a positive score loss. Repeated exact searches returned the same top-ten results for every tested query.

中文

平均分数损失与精确搜索发现的最高分数相比为0.0179, 而95%的启动间隔为0.0063-0.0357。18个查询中, 13个有积极分数损失。重复的精确搜索返回了每个测试查询的前十个结果。

English

Table 4.6: Exact-search and quota-limited search results Measure Quota-limited result Exact top result recovered 27.78% Mean recall of exact top ten 24.44% Mean candidate space examined 3.41% Mean score loss 0.0179 Queries with positive score loss 13 of 18 On the recorded CPU experiment, exact search had a median runtime of 0.474 seconds, a 95th-percentile runtime of 2.107 seconds and a maximum of 3.206 seconds. Quota-limited search over the same candidate space had a median runtime of 0.028 seconds.

中文

图4.6: 精确搜索和配额限制搜索结果测量配额限制搜索结果 精确最高结果恢复了 27.78% 精确前十的平均回忆 24.44% 平均候选空间检查了 3.41% 平均分数损失 0.0179 负分数损失的查询 13 在记录的CPU实验中, 精确搜索的平均运行时间为 0.474 秒, 95 个百分比运行时间为 2.107 秒, 最高为 3.206 秒。(原文数值: 0.028、18。)

English

Candidate availability was tested across four weather scenarios and nine styles, giving 36 combinations. Every tested combination retained all four logical clothing positions. In hot-weather cases, the outer-layer position was represented by the explicit masked No Outer Layer option.

中文

考生可用性在四种天气场景和九种风格中进行了测试, 提供了36种组合. 每种测试组合都保留了四个逻辑的服装位置. 在炎热天气的情况下, 外层位置由明确的掩盖无外层选项表示.

4.8 Summary

4.8 总结

English

The selected compatibility model achieved test AUC of 0.8583 and FITB accuracy of 0.6025 on the retained official subset. The three-seed comparison recorded a higher mean AUC for the proposed architecture than for the three controlled learned baselines. Removing the pairwise module produced a clear reduction, whereas the interval for removing clothing-position information crossed zero.

中文

选择的兼容性模型在保留的官方子集中达到0.8583的测试AUC和0.6025的FITB精度. 三种比较记录了拟议的架构的平均AUC比三个受控的学习基线高. 取消双模块显著减少, 而取消服装位置信息的间隔超过了零.

English

The selected model showed a substantial train-test gap but close validation and test performance. Validation-only calibration reduced negative log-likelihood and expected calibration error. The tested outfit-length weighting did not pass its deployment gate.

中文

选择的模型显示了大量的训练测试差距, 但验证和测试性能很接近. 仅通过验证定位降低了负面水平的预期定位错误. 测试的服装长度权重未经过其部署门口.

English

The product-independent candidate representation expanded the range of visible type-colour combinations. The quota-limited search recovered the same highest scoring result as exact search in 27.78% of the tested queries while requiring substantially less runtime. Chapter 5 examines the meaning and limitations of these results.

中文

产品独立的候选代表性扩大了可见的字体颜色组合范围. 限量搜索在测试的查询中占27.78%的精确搜索结果, 同时需要更少的运行时间. 第5章探讨了这些结果的含义和局限性.

Chapter 5: Analysis

第五章 分析

5.1 Introduction

5.1 引言

English

This chapter analyses what can reasonably be concluded from the results in Chapter 4. It considers the evidence provided by the controlled baselines, structural ablations, generalisation audit and candidate-search experiment. It also distinguishes results that support the proposed design from those that reveal uncertainty or limitations.

中文

本章分析了从第4章的结果中合理得出的结论, 它考虑了控制基线, 结构性消融, 泛化审计和候选搜索实验所提供的证据, 它还区分了支持拟议设计的结果与揭示不确定性或限制的结果。

English

The analysis remains within the scope of the experiments. A model score represents compatibility learned from the processed Polyvore data. It does not measure whether an individual user likes an outfit, whether the outfit is objectively fashionable or whether a recommended concept will lead to a purchase.

中文

分析仍然属于实验范围。模型分数代表了从处理的Polyvore数据中获取的兼容性。它不测量个人用户是否喜欢服装, 服装是否客观的时尚, 或者推荐的概念是否会导致购买。

5.2 Compatibility performance and baseline evidence

5.2 兼容性性能和基准证据

English

The selected compatibility model achieved an AUC of 0.8583 on the held-out test data. This means that the model usually assigned a higher score to a randomly selected positive outfit than to a randomly selected negative outfit (Fawcett, 2006). It does not mean that 85.83% of recommendations are correct or will be liked by users.

中文

选择的兼容性模型在持久测试数据上获得了0.8583的AUC。这意味着模型通常将比随机选择的负面服装更高的分数分配给了随机选择的积极服装 (Fawcett, 2006)。这并不意味着85.83%的推荐是正确的或会被用户喜欢。

English

The non-learned SigLIP similarity baseline reached an AUC of 0.7012. Frozen visual features therefore contained some information related to the dataset labels without compatibility training. The learned model nevertheless performed substantially better under the same processed test protocol. This difference supports the need to learn relationships among garments rather than relying only on general visual similarity, consistent with the distinction between similarity and complementarity made by McAuley et al. (2015).

中文

没有学习的SigLIP相似性基线达到0.7012的AUC。因此, 冻结的视觉功能包含了一些与数据集标签相关的信息, 而没有兼容性培训。然而, 在相同的加工测试协议下, 学习的模型表现得更好。这种差异支持学习服装之间的关系, 而不是仅依赖于一般的视觉相似性, 与McAuley等人 (2015) 所做的相似性和补充性之间的区别一致。

English

The proposed architecture also exceeded the three controlled learned baselines across the predefined seeds. Paired AUC intervals remained above zero for the comparisons with the bidirectional LSTM, type-aware model and set Transformer. Within this experiment, the result was therefore not produced by one favourable initialisation or one unusually weak baseline run.

中文

拟议的架构还超过了预定义的种子中三个控制学习基线。对对AUC间隔在与双向LSTM, 型号意识模型和集变压器的比较中保持在零以上。在本次实验中, 因此没有一个有利的初始化或一个异常弱的基线运行产生结果。

English

This does not demonstrate that the proposed model is universally superior to the published architectures. The comparison models were controlled adaptations using frozen SigLIP inputs and the project's four-position representation. The evidence shows that, under these shared conditions, the lightweight design was more effective than the selected alternatives. Direct comparison with published headline values would not be valid because the encoders, filtering and test coverage differ.

中文

这并不表明拟议的模型普遍优于公布的架构。比较模型是使用冻结的SigLIP输入和项目四位表示的控制适应。证据表明, 在这些共享条件下, 轻量级设计比选出的替代品更有效。直接与公布的标题值进行比较不会有效, 因为编码器, 过滤和测试覆盖范围不同。

English

The proposed model was also the fastest of the four learned architectures in the recorded training experiment. This supports its description as lightweight within the project. The timing remains dependent on the implementation, hardware and frozen-input pipeline and is not a universal speed comparison.

中文

拟议的模型也是记录的训练实验中学习的四个架构中最快的。这支持该模型在项目内被描述为轻量级的。时间仍然取决于实施, 硬件和冻结输入管道, 并不是普遍的速度比较。

5.3 Architectural evidence

5.3 建筑证据

English

Removing the pairwise module reduced mean FITB accuracy by 10.28 percentage points. The paired confidence interval remained entirely above zero. This provides the clearest structural evidence in the study: explicitly comparing garments contributed information that was not recovered by the overall outfit summary alone.

中文

通过删除双向模块, 平均FITB精度减少了10.28个百分点。对的信任间隔完全超过零。这提供了研究中最清楚的结构证据: 明确比较服装贡献的信息, 而仅仅在整体服装总结中无法恢复。

English

The finding accords with the relational treatment of compatibility in earlier work (Vasil-eva et al., 2018; Sarkar et al., 2023). A garment may be plausible in one combination but not another, so an average outfit representation can lose differences between par-ticular items. The pairwise component preserves a summary of these relationships without requiring a much larger attention model. This is evidence about the tested implementation rather than proof that pairwise aggregation is always preferable to attention.

中文

结果与早期作品中的相容性关系处理相符 (Vasil-eva et al., 2018; Sarkar et al., 2023)。一件服装在一个组合中可能是可信的,但不是另一个,因此平均服装表示可能会失去细节物品之间的差异。双重组件保留了这些关系的总结,而不需要更大的注意力模型。这是对测试的实施的证据,而不是证明双重聚合总是优先于注意力。

English

Removing clothing-position information produced only a small reduction, and its con-fidence interval crossed zero. The experiment therefore does not provide clear evid-ence that the position embedding improved FITB performance. One possible explana-tionisthatthevisualrepresentationandpairwisereleationsalreadycontaininformation that distinguishes broad garment roles. This is a possible explanation rather than a demonstrated cause.

中文

由于外衣位置信息的删除只造成了小幅减少,而其信心间隔超过了零。因此,实验没有提供明确的证据表明,嵌入位置改善了FITB性能。一个可能的解释是,视觉表现和双色球关系已经包含了区分广泛的服装角色的信息。这是一个可能的解释而不是一个被证明的原因。

English

The position embedding was retained because it adds little computational cost and gives the model an explicit indication of clothing role. Nevertheless, it should not be described as an empirically necessary component. A simpler model without it remains a reasonable alternative for future evaluation.

中文

由于它增加了较少的计算成本,并给模型提供了明确的服装角色,因此位置嵌入被保留了。然而,它不应该被描述为经验上必要的组成部分。没有它的更简单的模型仍然是未来评估的合理替代品。

5.4 Generalisation, overfitting and calibration

5.4 概括, 过度配件和校准

English

Training AUC of 0.9699 was considerably higher than test AUC of 0.8583. The differ-ence shows that the model fitted the training data more strongly than it generalised to unseen examples. The results therefore contain evidence of training-set overfitting.

中文

训练AUC为0.9699比测试AUC为0.8583要高得多。这种差异表明,模型比一般化为未见的例子更强地适应训练数据。因此,结果包含了训练集过拟合的证据。

English

Validation and test AUC differed by only 0.00145, and the saved checkpoint corresponded to the highest observed validation AUC. Validation-based checkpoint selection therefore behaved consistently on the held-out test split. It does not remove the train-test gap, but continuing beyond the selected epoch was not supported by the validation results. This is the limited role expected of early stopping: it controls continued fitting after validation performance peaks but does not guarantee the absence of overfitting (Prechelt, 1998).

中文

验证和测试AUC仅差0.00145, 而保存的检查点与最高的验证AUC相对应。因此, 基于验证的检查点选择在进行的测试分区上表现得一致。它不消除训练测试差距, 但继续超过所选择的时代并没有得到验证结果的支持。这是预期早期停止的有限作用: 它控制了验证性能峰值后的连续安装, 但不保证过度安装 (Prechelt, 1998)。

English

Further tuning against the current test results would weaken the independence of the test set. Additional regularisation experiments should therefore use validation-only decisions or introduce a new untouched evaluation split. The current checkpoint is a more defensible stopping point than repeatedly changing the model after inspecting its test performance.

中文

进一步调整与现行测试结果会削弱测试组的独立性。因此, 额外的规范实验应使用仅进行验证的决定或引入新的未接触的评估分区。目前的检查点比检测性能后重复改变模型更是可辩护的停留点。

English

Temperature scaling reduced validation negative log-likelihood and expected calibration error. The original scores were therefore more confident than necessary on the validation data. Calibration improves the interpretation of confidence but does not alter recommendation order or provide evidence of improved compatibility ranking (Guo et al., 2017).

中文

温度扩展降低了验证负日志概率和预期校准错误。因此, 原始分数对验证数据比必要更有信心。校准改善了对信任的解释, 但不会改变推荐顺序或提供改善兼容性排名的证据 (Guo et al., 2017)。

5.5 Outfit-length imbalance

5.5 服装长度失衡

English

The dataset was balanced between positive and negative labels, so the principal imbalance did not concern class counts. Instead, three-item outfits were more common than four-item outfits. Moderate length weighting produced a slightly higher overall mean AUC but reduced performance for the weakest length group and increased the difference between the two groups.

中文

数据集在正面和负面标签之间保持平衡, 因此主要的不平衡不涉及类数量。相反, 三项服装比四项服装更常见。中度长度权重产生了略高的总体平均AUC, 但降低了最弱的长度组的性能, 增加了两组之间的差异。

English

Selecting that configuration only because its overall AUC was marginally higher would conflict with the purpose of the weighting experiment. The predefined gate required the weakest group to improve rather than allowing a small overall gain to hide weaker group performance. Retaining the unweighted configuration was therefore consistent with the stated decision rule.

中文

选择这种配置仅因为其总体AUC略高, 就会与权重实验的目的相冲突。预定义的门槛要求最弱的组改进, 而不是允许小的总体收益来掩盖较弱的组性能。因此, 保留未权重的配置符合所述决策规则。

English

This result does not show that weighting can never improve the model. It shows only that the tested schemes did not provide a sufficiently balanced improvement under the present data and acceptance criteria.

中文

这一结果并不表明权重永远无法改善模型, 只表明, 根据现有的数据和接受标准, 测试的方案没有提供足够平衡的改善。

5.6 Relationship between learned scores and explicit constraints

5.6 学习得分与明确限制之间的关系

English

The application does not allow the compatibility score to decide every part of an outfit. This separation is necessary because statistical compatibility and feasibility answer different questions. The model estimates whether a complete combination resembles compatible relationships in the processed training data. It does not know that an outer layer may be unnecessary in hot weather, that a user-selected garment must remain fixed, or that a candidate belongs to the requested clothing range. These conditions are applied before or during search rather than learned indirectly from the score.

中文

应用程序不允许兼容性分数决定服装的每个部件。这种分离是必要的, 因为统计兼容性和可行性回答不同的问题。模型估计处理的训练数据中是否有完整的组合类似于兼容性关系。它不知道在炎热天气中外层可能是不必要的, 用户选择的服装必须保持固定, 或者候选属于所需的服装系列。这些条件是在搜索之前或搜索期间应用, 而不是间接从分数中学习。

English

The arrangement has two analytical advantages. First, a recommendation can be traced through candidate construction, constraint filtering and final scoring. If no outer layer is shown, the system can identify a weather mask rather than attribute the decision vaguely to the neural model. If an unsuitable clothing range appears, the error lies in filtering or metadata rather than in the compatibility score. This is a limited form of procedural explanation: it does not explain the visual features learned by SigLIP, but it identifies which system component made a deterministic decision.

中文

系统可以通过候选构建, 限制过和最终分数来追踪推荐。如果没有显示外部层, 系统可以识别天气面具, 而不是模糊地归因于神经模型。如果出现不合适的服装范围, 错误在于过或元数据而不是兼容性分数。这是一个有限的程序解释形式: 它不解释SigLIP学习的视觉特征, 但它确定了哪个系统组件做出决定性决定。

English

Second, the rules stop the model from being evaluated for knowledge that its training objective did not provide. A positive or negative outfit label does not directly encode local weather, personal wardrobe ownership or application state. Expecting the compatibility head to infer these factors would mix separate sources of evidence and make failures harder to diagnose. The modular design therefore supports auditability even though it may be less flexible than one end-to-end model.

中文

根据规则, 模型无法根据其培训目标提供的知识进行评估。一个积极或负面的服装标签不会直接编码当地天气, 个人衣柜所有权或应用状态。期望兼容性负责人推断这些因素将混合不同的证据来源, 使故障难以诊断。因此, 模块化设计支持可审核性, 尽管它可能比一个端到端模型更不灵活。

English

This division also creates limitations. A rule can exclude a combination that the model would score highly, and errors in garment metadata can remove valid candidates before ranking.

The reported model AUC does not test those rule failures. For this reason, model evaluation and 36-case candidate-availability testing were reported separately. The deterministic fallback is handled in the same way. It keeps the application usable when model artefacts are absent, but its output should not be combined with learned-model results or described as equivalent in quality.

中文

这种分类也会造成限制。一个规则可以排除模型会得分高的组合, 服装元数据中的错误可以提前排名有效的候选。报告模型AUC不测试这些失败。因此, 模型评估和36例候选可用性测试分别报告。确定性倒退是以同样的方式处理的。它使模型在文物缺失时应用程序可用, 但其输出不应与学习模型结果结合, 或被描述为质量相当。

5.7 Candidate-search completeness and latency

5.7 申请人搜索的完整性和延迟

English

Quota-limited search examined an average of 3.41% of the feasible combinations. It recovered the same highest-scoring result as exact search in 27.78% of 18 queries and recovered an average of 24.44% of the exact top ten. The quota stage therefore frequently removed combinations that the compatibility model would otherwise have ranked highly.

中文

配额有限的搜索平均检查了可行的组合的3.41%。在18个查询中, 它获得了27.78%的精确搜索和24.44%的精确前十个查询中相同的最高分数结果。因此, 配额阶段经常删除了兼容性模型否则将排名高的组合。

English

A positive score loss in 13 of the 18 queries confirms that the change was not limited to rearranging equally scored candidates. Score loss refers only to the deployed model objective, however; it does not establish that a person would prefer the exact-search result.

中文

在18个查询中13个中出现了积极的分数损失, 证实了这一变化并非仅限于重新排列同等分数的候选。

English

Exact search increased median runtime from approximately 0.028 seconds to 0.474 seconds on the recorded CPU experiment. Its 95th-percentile runtime remained close to two seconds. The experiment identifies a clear trade-off: exact search provides completeness under the model objective, while quota search provides lower latency. This reflects the broader retrieval trade-off between search coverage and computational cost (Johnson et al., 2021).

中文

精确搜索在记录的CPU实验中增加了平均运行时间从大约0.028秒到0.474秒。其95个百分比运行时间保持在近两秒的时间内。实验确定了明确的折衷：精确搜索在模型目标下提供完整性，而配额搜索提供较低延迟。这反映了搜索覆盖和计算成本之间的更广泛的检索折衷（Johnson等人，2021）。（原文引用：（Johnson et al., 2021）。）

English

Within the tested candidate sizes, the additional runtime remained compatible with an interactive prototype. This statement applies only to the recorded hardware and bounded search space. A substantially larger catalogue could require approximate retrieval, staged ranking or another search strategy.

中文

在测试的候选尺寸中，额外的运行时间仍然与互动原型兼容。这项声明仅适用于记录的硬件和有限的搜索空间。一个更大的目录可能需要近似的检索，阶段排名或其他搜索策略。

5.8 Coverage, domain shift and bias limitations

5.8 覆盖范围，域名转移和偏见限制

English

Only 29.50% of the official FITB questions could be represented by the four supported clothing positions. The resulting FITB accuracy is valid only for the retained subset and cannot be presented as performance on the complete official benchmark.

中文

只有29.50%的官方FITB问题可以由四个支持的服装位置来代表。结果的FITB准确性仅适用于保留的子集，不能被呈现为完整的官方基准值的性能。

English

Menswear evidence was particularly sparse. Only 46 retained FITB questions were labelled as menswear, and the training data contained very few purely menswear outfits. The figures identify a coverage problem but do not estimate a dependable performance difference between clothing ranges.

中文

调查结果显示，只有46个受访者被FITB评为男性服装，而培训数据中很少有纯男性服装。数据显示，覆盖问题存在，但没有估计服装范围之间的可靠性能差异。

English

The feature-domain classifier separated Polyvore and Zara items with an AUC of 1.000. Their frozen feature distributions were therefore distinguishable under the diagnostic procedure. This does not measure how much compatibility accuracy would decline on Zara because compatible Zara outfit labels were unavailable. The result instead prevents an unsupported assumption that performance transfers unchanged between the two sources.

中文

功能域分类器分离了Polyvore和Zara的产品，AUC为1.000。因此，它们的冷功能分布在诊断程序下可以区分。这并不能衡量Zara的兼容度准确度会下降多大，因为兼容的Zara服装标签无法使用。结果相反，防止了不支持的假设，即两种来源之间的性能转移没有变化。（原文数值：1.000。）

English

The candidate system avoids presenting historical product brands and allows the user to choose a clothing range, retain the uploaded garment and reject alternatives. These choices improve control and reduce some immediate product risks, but they do not re-move bias inherited from Polyvore labels or the frozen visual representation (Mehrabian et al., 2021; Selbst et al., 2019).

中文

候选系统避免展示历史产品品牌, 并允许用户选择服装系列, 保留上传的服装和项目替代品. 这些选择改善了控制和减少了一些直接的产品风险, 但它们不会重新移动从Polyvore标签或冻结的视觉表示中继承的偏见 (Mehrabian等人, 2021; Selbst等人, 2019). (原文引用: (Mehrabian et al., 2021; Selbst et al., 2019).)

English

Finally, no participant evidence is available. The project can support claims about offline compatibility discrimination, candidate completion, search completeness and system behaviour. It cannot establish emotional benefit, trust, usability across populations or preference improvement.

中文

最后, 没有参与者证据. 该项目可以支持关于离线兼容性歧视, 候选完成, 搜索完整性和系统行为的说法. 它不能建立情感效益, 信任, 跨人口的可用性或偏好改善.

5.9 Summary

5.9 总结

English

The strongest model evidence comes from the controlled architecture comparison and removal of the pairwise module. Together, these results support a lightweight relational head on frozen visual representations within the processed Polyvore task. Evidence for the clothing-position embedding is inconclusive.

中文

最强的模型证据来自于对照架构比较和对模块的移除. 这些结果一起支持了加工的多体任务中冻结的视觉表示的轻量关系头. 服装位置嵌入的证据是不确定性的.

English

The selected checkpoint retains an overfitting tendency, although validation-based selection produced closely aligned validation and test results. Calibration improved validation confidence measures, while the tested length-weighting schemes did not provide a sufficiently balanced improvement.

中文

选择的检查点仍然存在过拟合的趋势, 尽管基于验证的选择产生了紧密一致的验证和测试结果. 校准改进了验证的可靠度, 而测试的长度权重方案没有提供足够平衡的改善.

English

Exact search substantially improved completeness under the model score at the cost of increased runtime. The evidence remains bounded by restricted FITB coverage, sparse menswear data, measurable feature-domain separation and the absence of participant evaluation. Chapter 6 uses these positive and negative findings to answer the research questions and assess the contribution.

中文

精确搜索在模型分数下显著提高了完整性, 以牺牲增加运行时间. 证据仍然受到FITB的限制, 男性服装数据稀缺, 可测量的特征域分离和参与者评估的缺失. 第6章使用这些积极和负面的发现来回答研究问题并评估贡献.

Chapter 6: Discussion

第六章 讨论

6.1 Introduction

6.1 引言

English

This chapter brings together the literature, experimental findings and system evidence to answer the research questions. The discussion separates three levels of claim. The first concerns discrimination and completion ranking on the processed Polyvore data. The second concerns the contribution of model components under controlled ablation. The third concerns the behaviour of the complete prototype, including candidate search, constraints and fallback. Keeping these levels separate is important because offline compatibility accuracy does not demonstrate that users will prefer, trust or purchase the resulting outfits.

中文

这一章将文献、实验发现和系统证据汇集起来，以回答研究问题。讨论分开了三种要求层次。第一是关于处理的 Polyvore 数据的歧视和完成排名。第二是关于模型组件在控制式消融下所做出的贡献。第三是关于完整原型的行为，包括候选搜索、限制和倒退。保持这些层次的分离是重要的，因为离线兼容度不表明用户会更喜欢、信任或购买结果的服装。

6.2 Relationship to previous compatibility research

6.2 与以前的兼容性研究的关系

English

Previous studies have represented an outfit as a sequence, as type-specific pair relationships, or as a set processed by attention (Han et al., 2017; Vasileva et al., 2018; Sarkar et al., 2023). The present results support the premise shared by these approaches: compatibility is not adequately represented by isolated visual similarity. Mean pairwise SigLIP cosine similarity reached an AUC of 0.7012 in the fixed-seed comparison, whereas the learned relational head reached 0.8573 in the same experiment. The mean-pooled logistic baseline was weaker still, with test AUC of 0.5532 on the disjoint data. Frozen visual features therefore contained relevant information, but averaging them did not express the relationships required by the task.

中文

之前的研究已经将一个服装作为一个序列，作为类型特定的对关系，或作为一个由注意处理的集合（Han et al., 2017; Vasileva et al., 2018; Sarkar et al., 2023）。目前的结果支持这些AP方法共享的前提：兼容性不充分通过孤立的视觉相似性表示。平均对 SigLIP 共性相似性在固定种子比较中达到0.7012的AUC，而学到的关系头在同一实验中达到0.8573。平均合并物流基线仍然较弱，测试AUC在分离数据上为0.5532。因此，冻结的视觉特征包含相关信息，但平均它们并不能表达任务所需的关系。

English

Across three seeds, the proposed model produced mean AUC of 0.8596, above the controlled bidirectional LSTM, type-aware pairwise and set-Transformer implementations. This is evidence for the selected lightweight design, but it does not show that it outperforms the original published systems. The comparison models were adapted to the same four-position tensors, frozen embeddings, processed labels and training conditions. They compare architecture families while controlling the surrounding pipeline. Published headline scores use different inputs, splits, negative construction and implementation details, so a direct numerical ranking against those results would not be valid. The supported conclusion is narrower: under this project's protocol, the small relational head was more effective and stable than the three controlled alternatives.

中文

在三个种子中, 拟议的模型产生的平均AUC为0.8596, 高于控制的双向LSTM, 类型感知对和设置变压器实现。这是选定的轻量级设计的证据, 但它并没有显示出它比较原始发布的系统。比较模型适应相同的四位置子, 结嵌件, 加工标签和训练条件。它们在控制周围的管道时比较建筑家族。发表的标题分数使用不同的输入, 分区, 负面构建和实施细节, 因此对这些结果进行直接数值排名将不有效。支持的结论更窄: 根据该项目的协议, 小关系头比三个控制替代品更有效和稳定。

English

The frozen SigLIP encoder also affects how the result should be interpreted. SigLIP was previously trained on large-scale image-text data and supplied a transferable representation (Zhai et al., 2023). This project did not reproduce or fine-tune that representation learning. Its original training cost and learned biases remain part of the dependency, even though the project-specific head has approximately 379,000 trainable parameters and trains quickly. The short head-training time is consistent with reusing precomputed embeddings; it is not the total computational cost of creating the visual representation from first principles.

中文

冻结的SigLIP编码器还影响了结果应该如何解释。SigLIP以前在大规模的图像文本数据上训练并提供了可转移的表示 (Zhai et al., 2023)。该项目没有复制或细调表示学习。其原始的培训成本和学习偏见仍然是依赖的一部分, 尽管项目特定的头部约有379,000个可训练的参数, 并快速训练。短的头部训练时间与重复使用预计算嵌入一致; 它不是从第一原则创建视觉表示的总计算成本。

6.3 Answer to RQ1: comparative performance

6.3 回答RQ1: 比较性能

English

RQ1 asked how the proposed model compared with non-learned and controlled learned baselines under consistent protocols. The evidence gives a positive but bounded answer. The selected checkpoint achieved test AUC of 0.8583, with a 95% confidence interval of 0.8524-0.8641. It was clearly above deterministic random scores and mean pairwise cosine similarity. Across three seeds, its mean AUC exceeded each controlled learned architecture; paired bootstrap intervals for the AUC differences did not include zero. Its standard deviation of 0.0009 was also lower than those of the comparison models.

中文

RQ1 问, 该模型与不学习和控制的学习基线相比如何。证据提供了积极但有限的答案。选择的检查点达到0.8583的测试AUC, 95%的可靠性间隔为0.8524-0.8641。它显然超过了确定性随机分数和平均对方共数相似性。在三个种子中, 其平均AUC超过了每个受控制的学习架构; 对AUC差异的对联启动间隔不包括零。其标准偏差为0.0009也低于比较模型。

English

These results indicate that the model learned a repeatable distinction between observed and synthetically corrupted outfits within the processed disjoint setting. AUC measures the probability that a randomly selected positive example is ranked above a randomly selected negative example (Fawcett, 2006). It does not mean that 85.83% of proposed outfits would be judged fashionable, nor does it measure individual taste. Using same-category replacements makes the negative task more demanding than arbitrary cross-category corruption, but compatibility is still defined by the available data and negative generator.

中文

这些结果表明, 模型在处理的分离设置中学到了使用过的服装和合成腐败的服装之间的重复区别. AUC测量了随机选择的积极例子在随机选择的负面例子上排名的可能性 (Fawcett, 2006). 这并不意味着85.83%的推荐服装将被认为是时尚的, 也不测量个人口味. 使用相同类别的替代品使得负面任务比任意跨类别腐败更具要求, 但兼容性仍然由可用的数据和生成器负面定义.

English

FITB accuracy of 0.6025 provides a second form of evidence because the model must select one answer from four alternatives. The result is above the chance level of 0.25, and its confidence interval does not approach chance. However, only 29.50% of the official questions were retained. It therefore applies to the four supported clothing positions with available embeddings, not to the complete Polyvore disjoint benchmark. Together, AUC and FITB demonstrate useful offline discrimination and completion ranking, while the coverage figure limits the scope of the latter result.

中文

根据FITB的准确性为0.6025提供了第二种证据, 因为模型必须从四种替代方案中选择一个答案. 结果高于0.25的机会水平, 其信心间隔不接近机会. 然而, 仅有29.50%的官方问题被保留. 因此, 它适用于可用的嵌入式支持的四个服装位置, 而不是全 Polyvore分离基准. AUC和FITB一起证明了有用的离线歧视和完成排名, 而覆盖率数字限制了后者的范围.

6.4 Answer to RQ2: architecture and training choices

6.4 答案RQ2: 建筑和培训选择

English

RQ2 asked which design and training choices were supported by the combined evidence. The strongest structural result concerns explicit pairwise comparison. Removing the pairwise module reduced mean FITB accuracy by 10.28 percentage points, with a paired 95% interval from 8.96 to 11.57 points. The corresponding classification AUC also fell. This supports the view that compatibility depends on interactions among garments rather than only an outfit-level average.

中文

二问了合并证据支持哪些设计和培训选择. 较长的结构性结果涉及对比性. 取消对模块减少了平均FITB准确度10.28个百分点, 对比的95%间隔从8.96到11.57个点. 相应的分类AUC也下降. 这支持认为兼容性取决于服装之间的相互作用而不是仅在服装水平的平均水平.

English

The evidence for clothing-position embeddings was weaker. Removing them reduced mean FITB accuracy by 0.47 percentage points, but the paired interval ranged from -0.13 to 1.06 points. Since the interval crosses zero, the experiment does not establish a reliable improvement. Retaining the component is defensible because its cost is small and its semantic role is clear, but it should not be described as necessary. A larger or more position-diverse evaluation would be needed to resolve the effect.

中文

服装位置嵌入的证据较弱。删除它们使平均FITB精度降低0.47个百分点,但对间隔范围从0.13到1.06个百分点。从间隔跨界,实验没有确立可靠的改善。保留组件是可辩护的,因为其成本较小,其语义作用很清楚,但不应该被描述为必要。解决效果需要更大或更大的位置差异评估。

English

The selected model was trained with binary cross-entropy on positive and negative outfits, then used as a scorer for candidate ranking. A scalar compatibility estimate can order candidates, so this use is coherent, but it is not equivalent to directly optimising FITB or user preference. A tested pairwise ranking objective did not improve the retained evaluation and was not selected. This negative result is informative: changing the loss alone did not overcome limitations in candidate construction and representation.

中文

选择的模型采用对积极和负面的外观进行双向交叉熵训练,然后作为候选排名的得分符。一个规模兼容性估计可以命令候选,因此这种使用是一致的,但它并不是直接优化FITB或用户偏好的等价。测试的双向排名目标没有改善保留的评估,并没有被选中。这种负面结果是信息性的:仅仅改变损失无法克服候选构建和表示的限制。

English

Generalisation evidence is mixed. Validation and test AUC differed by only 0.00145, supporting the use of validation performance for checkpoint selection. By contrast, training AUC exceeded test AUC by 0.1116, and validation performance declined after epoch 6. The model therefore overfit the training distribution, although early stop-ping selected a checkpoint that transferred consistently to the held-out test split. Early stopping controlled the observed deterioration; it did not eliminate overfitting (Prechelt, 1998).

中文

综合性证据是混合的。验证和测试AUC仅差0.00145,支持使用验证性能进行检查点选择。相比之下,训练AUC超过测试AUC0.1116,并且验证性能在6期后下降。因此,该模型对训练分布进行了重叠,尽管早期停止选择了一个检查点,并将其连续转移到持续的测试分区。早期停止控制了观察到的恶化;它并没有消除过度装 (Prechelt, 1998)。

English

Temperature scaling reduced validation negative log-likelihood and expected calibration error without changing candidate ordering, consistent with its role as a post-hoc calibration method (Guo et al., 2017). These improvements apply to the validation distribution and do not convert compatibility scores into probabilities of human approval. Similarly, outfit-length weighting was rejected because a small gain in overall AUC coincided with weaker performance for the lowest-performing length group and a larger group gap. The decision follows the predefined acceptance gate rather than establishing that weighting is generally ineffective.

中文

温度扩展减少了验证负记录概率和预期校准错误,而不改变候选排序,与其作为后期校准方法的作用一致 (Guo et al., 2017)。这些改进适用于验证分布,并不会将兼容性分数转换为人体验证的概率。同样,服装长度权重被拒绝,因为AUC总体的小幅增长与最低性能长度组的性能和更大的组差距相吻合。该决定遵循预定义的接受门,而不是确定权重通常是无效的。

6.5 Answer to RQ3: candidate search and system control

6.5 回答RQ3: 候选搜索和系统控制

English

RQ3 concerned the trade-off between quota-limited retrieval and exact constrained search over abstract garment states. In the 18-query comparison, the quota path examined only 3.41% of combinations on average. This reduced median CPU time from 0.474 seconds for exact search to 0.028 seconds, but it recovered the exact highest-scoring result in only 27.78% of queries and achieved 24.44% mean recall of the exact top ten. Mean score loss was positive, with a bootstrap interval that did not include zero.

中文

在18个查询比较中, 配额路径平均只检查了3.41%的组合。这将精确搜索的CPU时间从0.474秒缩短到0.028秒, 但在仅27.78%的查询中恢复了最高分数的结果, 并实现了24.44%的精确回忆。平均分数损失是积极的, 启动间隔不包括零。

English

For the measured candidate space, exact enumeration therefore improved complete-ness according to the deployed model objective, while retaining a median runtime below half a second and a maximum of 3.206 seconds. This supports its use in the current prototype. It does not show that exact search will remain practical for an unrestricted catalogue. The number of complete combinations grows multiplicatively with the candidates in each position, so later expansion may require pruning or ap-proximate retrieval. Techniques for large-scale similarity search are well established (Johnson et al., 2021), but adopting one would create a new recall-latency trade-off that should be measured rather than assumed.

中文

对于测量候选空间, 精确编译因此根据部署的模型目标提高了完整性, 同时保持了平均运行时间在半秒以下, 最高为3.206秒。这支持其在当前原型中使用。这并不表明对无限制目录保持了精确搜索实用性。每位候选的完整组合数量增长多倍, 因此后期扩展可能需要剪裁或近似检索。大规模相似性搜索的技术已经很好地建立 (Johnson等人, 2021), 但采用一种将创造出新的回忆延迟交易, 而不是假设。(原文引用: (Johnson et al., 2021)。)

English

The abstract candidate representation also changed the role of recommendation. Polyvore did not become a catalogue from which users were instructed to select archived products. Instead, evidence from the training split was aggregated into product-independent combinations of clothing position, broad type, colour, style and clothing range. This matches the interface, which displays a coloured icon unless an item belongs to the user's local wardrobe. Compared with the earlier retailer-derived space, the artefact increased visible type-colour combinations from 122 to 271. This shows broader representational coverage; it does not prove that every combination is available in shops or will be preferred by a user.

中文

抽象候选表示也改变了推的作用。Polyvore并没有成为用户指示选择存档产品的目录。相反, 从培训分类的证据被汇集产品独立的服装位置, 宽型, 颜色, 风格和服装范围的组合。这与界面相匹配, 显示了彩色图标, 除非一个项目属于用户的当地衣柜。与早期零售商衍生空间相比, 该文物增加了可见的字体颜色组合从122到271。这显示了更广泛的表示覆盖范围; 这并不证明每个组合都可在商店中使用或被用户喜欢。

English

Search remained subordinate to explicit feasibility rules. The user's selected item was fixed, weather logic could mask the outer-layer position, and clothing range was selected manually. All 36 weather-style coverage cases retained the four logical positions. These checks support functional completeness within the defined representation. They do not assess whether every recommendation is seasonally or culturally appropriate, and the deterministic fallback ensures continuity rather than stylistic correctness.

中文

搜索仍然依赖于明确的可行性规则。用户选择的项目是固定的，天气逻辑可以掩盖外层位置，服装范围是手动选择的。所有36种天气风格覆盖案例都保留了四个逻辑位置。这些检查支持在定义的表现中功能完整性。它们不评估每个推荐是否是季节性或文化性适当的，确定性后退确保连续性而不是风格正确性。

6.6 Achievement of the research objectives

6.6 实现研究目标

English

The seven objectives were achieved to different degrees. O1 was addressed by the critical review of fashion compatibility, transferable visual representations, evaluation, search and responsible recommendation. O2 was achieved within the approved secondary-data scope. O3 was achieved through the frozen-SigLIP compatibility pipeline and saved lightweight head.

中文

通过批判性综述时尚兼容性，可转移视觉表示，评估，搜索和负责任推荐，O1得到了实现。O2在经过验证的二次数据范围内实现。O3通过冻结SigLIP兼容性管道实现，并节省了轻量级头部。

English

O4 was achieved through random and cosine controls, a mean-pooled logistic model, and controlled recurrent, type-aware and attention-based models. These are architecture-family comparisons rather than exact reproductions. O5 was substantially achieved through discrimination metrics, FITB, calibration, ablation, three seeds, confidence intervals, overfitting, length imbalance, runtime and domain diagnostics. It was not achieved for demographic fairness or user-centred outcomes because those variables were neither available nor collected. O6 was achieved within the bounded prototype through abstract candidates, exact search, explicit constraints and fallback. O7 is addressed by distinguishing demonstrated system behaviour from unsupported claims about users and markets.

中文

通过随机和共数控制，平均聚合物流模型和控制的复发型，类型感知和关注型模型来实现O5。这些模型是建筑-家庭比较而不是精确的复制。O5基本上通过歧视指标，FITB，校准，消融，三种种子，信任间隔，过度匹配，长度失衡，运行时间和域诊断实现。它并没有实现人口公平或用户中心的结果，因为这些变量既不存在，也没有收集。O6通过抽象候选，精确搜索，明确的限制和倒退实现在有限原型中。解决O7的问题是通过区分已证明的系统行为与有关用户和市场的未得到支持的说法。

English

These objectives support completion of the stated investigation, but not every problem in fashion recommendation. Benchmark coverage, menswear representation, target-domain validation and human evaluation remain partial or untested. Reporting them in this way preserves the connection between the research plan and the evidence produced.

中文

这些目标支持完善的研究，但时尚推中存在许多问题。标准标志覆盖，男装代表，目标领域验证和人体评估仍然是部分或未经测试的。这样的报告使研究计划与产生的证据保持联系。

6.7 Validity and confidence in the findings

6. 7 结果的有效性和可靠性

English

Construct validity is limited by the operational definition of compatibility. The model learned from observed Polyvore outfits and same-category replacements. These labels are reproducible, but they cannot capture every reason why a person might combine garments. FITB is closer to candidate completion than binary classification, yet it remains an offline choice among four supplied answers. Confidence should therefore be placed in dataset-relative ranking, not in a universal judgement of style.

中文

构建有效性由兼容性的运用定义限制。从观察到的多体服装和同类替代品中学到的模型。这些实验室可以复制，但它们不能捕捉一个人为什么可以组合服装的每一个原因。FITB比二进制分类更接近候选完成，但它仍然是提供的四个答案中离线选择。因此，应该在数据集相关排名中置信，而不是在普遍的风格判断中。

English

Internal validity was strengthened by common splits, frozen features, fixed comparison settings, validation-based checkpointing and paired analysis. The disjoint split reduces product memorisation across the main divisions. Nevertheless, the comparison architectures were adaptations, and implementation choices may favour some models. Three seeds reveal part of the training variation but do not characterise all hyperparameter uncertainty. Bootstrap intervals describe uncertainty conditional on the retained samples and scoring procedure; they do not correct sampling bias in the original data (Efron and Tibshirani, 1993; Bouthillier et al., 2021).

中文

由于这种情况，我们可以使用“分类”的方法来进行测试，并将其进行测试，并将其进行测试。由于分类的特性，我们可以使用“分类”的方法来测试，并将其进行测试。由于分类的特性，我们可以使用“分类”的方法来测试，并将其进行测试。由于分类的特性，我们可以使用“分类”的方法来测试，并将其进行测试。由于分类的特性，我们可以使用“分类”的方法来测试，并将其进行测试。由于分类的特性，我们可以使用“分类”的方法来测试，并将其进行测试。由于分类的特性，我们可以使用“分类”的方法来测试，并将其进行测试。由于分类的特性，我们可以使用“分类”的方法来测试，并将其进行测试。（原文引用：(Efron and Tibshirani, 1993; Bouthillier et al., 2021)。）

English

External validity is more restricted. The feature-domain classifier separated Polyvore and Zara embeddings with AUC 1.000. This shows that the two sampled sources were readily distinguishable in the chosen representation, but it does not quantify how much compatibility performance would change on new retail or user photographs. The four-position representation excludes dresses and other multi-position garments, and only 29.50% of official FITB questions were evaluable. Menswear evidence is particularly sparse. The findings should therefore be generalised only to similar processed data and supported garment structures.

中文

外部有效性较有限。特征域分类器将Polyvore和Zara嵌入式产品分开为1000AUC。这表明，两种采样来源在选择中表示中很容易区分，但它并未确定新零售或用户照片的兼容性性能将会有多大的变化。四个位置的表示不包括服装和其他多位置的服装，只有29.50%的官方FITB问题可评估。男子服装证据特别稀少。因此，该发现应仅将其概括到类似的处理数据和支持的服装结构。（原文数值：1.000。）

English

Ecological validity is also untested because no participant study was conducted. The application demonstrates that its components can be integrated and that tested actions function; it does not demonstrate usefulness in daily life. This distinction follows the broader finding that recommender evaluation requires metrics matched to the intended task and that offline accuracy alone is incomplete (Herlocker et al., 2004; Deldjoo et al., 2023).

中文

由于没有参与者研究进行,生态有效性也没有被测试。应用程序证明其组件可以集成,测试的操作功能;它不证明在日常生活中有用。这种区分是因为更广泛的发现,推评估需要与预期任务匹配的指标,而离线准确性本身是不完整的(Herlocker et al., 2004; Deldjoo et al., 2023)。(原文引用: (Herlocker et al., 2004; Deldjoo et al., 2023)。)

6.8 Contribution, benefit and responsible use

6.8 贡献,福利和责任使用

English

The principal contribution is not a new foundation model. It is an auditable combination of reused visual representations, a small relational compatibility head, controlled evaluation and constrained abstract search. The implementation shows how a resource-limited project can investigate compatibility without training a large image encoder end to end. Separating learned scoring from feasibility rules also makes it possible to identify whether a failure arose from model ranking, candidate availability or a contextual constraint.

中文

主要贡献不是一个新的基础模型。它是重复使用的视觉表示,一个小的关系兼容性头,控制评估和有限的抽象搜索的可审核组合。实施表明,资源有限的项目如何在不训练一个大型图像编码器的最后进行兼容性调查。从可行性规则中分离学到的分数也可以确定是否出现失败是由于模型排名,候选可用性或文本上的限制。

English

Product-independent output provides a practical design benefit. A recommendation can describe a broad type and colour without suggesting that one brand contains the only suitable product. The selected garment remains visible and fixed, while abstract icons distinguish generated suggestions from garments saved by the user. Local ward-robe storage and the absence of automatic gender inference are consistent with the prototype's privacy boundary. These are implemented properties, not measured user benefits.

中文

产品独立的输出提供了实用设计效益。推荐可以描述一个广泛的类型和颜色,而不推荐一个品牌包含唯一合适的产品。所选的服装仍然可见和固定,而抽象图标则区分生成的推荐与用户保存的服装。本地衣柜存储和缺乏自动性推断与原型的隐私界限一致。这些是实现的属性,而不是测量用户的好处。

English

Industrial or commercial value remains prospective. The system could support ward-robe organisation, style exploration or later links to retailers and independent designers, but conversion, retention, willingness to pay and designer outcomes were not measured. Deployment would also require current product data, verified image permissions, accessibility work, security review, service monitoring and evidence across clothing ranges and styles. The dissertation can therefore present a working and extensible research artefact, not a validated business.

中文

工业或商业价值仍然是前景的。该系统可以支持衣柜组织,风格探索或与零售商和独立设计师的后续联系,但转换,保留,付费愿意和设计结果没有被测量。部署还需要当前的产品数据,验证的图像许可,可访问性工作,安全综述,服装范围和风格的服务监测和证据。因此,论文可以呈现一个工作和可扩展的研究文物,而不是一个验证的业务。

6.9 Implications for applied AI practice

6.9 对应用人工智能实践的影响

English

The results suggest that model scale should follow the problem rather than be treated as the main indication of technical quality. The proposed head trained faster than the controlled recurrent model and used a similar or smaller number of trainable parameters than the neural alternatives. Its stronger result came from an architecture that matched the relational task, especially the explicit pairwise summary. This does not establish that small models are generally more accurate. It shows that, once a transferable representation is available, a focused downstream component can be sufficient for a bounded application.

中文

结果表明, 模型规模应该遵循问题, 而不是被视为技术质量的主要指标. 拟议的头脑训练比控制的复制模型更快, 使用了类似或更少的可训练参数比神经替代方案. 它的更强大的结果来自一个与关系任务, 特别是明确的双重总结相匹配的架构. 这并不表明小型模型一般更准确. 这表明, 一旦可转移的表示可用, 一个集中下游组件可以为有限的应用而足够.

English

The development process also shows the value of treating data transformation as part of the model. The four-position mapping, construction of negatives, FITB eligibility rules and clothing-range labels determine which relationships the model can learn and which cases can be evaluated. These choices are not neutral preparation steps. For example, the exclusion of multi-position garments directly caused part of the FITB coverage loss, while sparse menswear-labelled data restricted subgroup conclusions. An applied system should therefore version its data mapping and report coverage together with performance, rather than presenting only the final checkpoint.

中文

开发过程还显示了处理数据转型的价值。四位位置映射, 负面的构建, FITB资格规则和服装范围标签决定了模型可以学习哪些关系, 哪些情况可以评估。这些选择不是中立的准备步骤。例如, 排除多位服装直接导致了FITB覆盖损失的一部分, 而稀少的男性服装标记数据限制了子组结论。因此, 应用系统应该将其数据映射和报告覆盖情况与性能一起编辑, 而不是仅呈现最终检查点。

English

The same principle applies to candidate generation. The learned score did not produce useful recommendations until possible garment states were represented, filtered and searched. The exact-search experiment showed that an apparently efficient quota could change the result before the model had an opportunity to score most combinations. In a larger system, retrieval quality would therefore need its own acceptance criteria. Improving the ranker while leaving an unmeasured retrieval bottleneck could produce no visible improvement for the user.

中文

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English

Finally, auditability came from ordinary engineering controls rather than from one explanation method. Saved configurations, fixed seeds, hashes, separate evaluation outputs, explicit constraints, failure-aware loading and a rule fallback make the path from evidence to interface more inspectable. They do not make the neural representation fully interpretable, but they reduce ambiguity about what was trained, selected and deployed. For an MSc Applied AI project, this system-level traceability is part of technical competence because it connects an experimental result to reproducible software behaviour.

中文

最后,可审核性来自于普通的工程控制,而不是一个解释方法。保存的配置,固定种子,哈希,单独的评估输出,明确的限制,故障意识的加载和规则倒退使证据到界面的路径更可检查。它们不会使神经表达完全可解释,但它们会减少对训练,选择和部署的模糊性。对于MSc应用人工智能项目来说,这种系统级追溯性是技术能力的一部分,因为它将实验结果与可复制的软件行为联系起来。

6.10 Prioritised future work

6.10 优先工作

English

Future work should address evidence gaps before adding complexity. The first priority is licensed data expansion. A new dataset should include broader menswear coverage, documented provenance and explicit structures for dresses and other one-piece garments. This would require data mapping, new position occupancy rules, regenerated frozen embeddings, retraining and repetition of the AUC, FITB, ablation and subgroup evaluation. Dresses should not simply be relabeled as inner tops because that would incorrectly permit a separate bottom.

中文

未来的工作应该在增加复杂性之前解决证据缺口。第一优先事项是许可证数据扩展。新数据集应包括更广泛的男性服装覆盖范围,记录来源和连衣裙和其他单件服装的明确结构。这需要数据映射,新的位置占用规则,再生的结嵌入,重新训练和重复AUC, FITB, 除和子组评估。服装不应该简单地重新命名为内顶,因为这会错误地允许单独的底部。

English

The second priority is target-domain validation. A modest, legally reusable set of complete outfits from a contemporary target domain should be labelled under a documented protocol. It could test whether compatibility ordering transfers beyond Polyvore and whether calibration remains valid. If performance falls, carefully bounded encoder adaptation or domain-robust training could be compared with the frozen baseline. This work requires additional computing time and permission review, but reliable data are the main requirement.

中文

目标域验证是第二个优先事项。一个适度的,法律上可重复使用的现代目标域的完整穿搭集应在文档协议下标记。它可以测试是否兼容性命令转移超越Polyvore,并是否校准仍然有效。如果性能下降,可以与结的基线进行仔细限制的编码器适应或域强大的培训。此工作需要额外的计算时间和许可综述,但可靠数据是主要的要求。

English

The third priority is participant evaluation after ethics approval. A study could compare recommendations from the learned model, the rule fallback and a controlled baseline while measuring perceived usefulness, choice diversity and trust. Recruitment, in-formed consent, withdrawal, secure data handling and a defined analysis plan would be required. Until such a study is completed, interface completeness cannot be used as evidence of satisfaction.

中文

第三个优先事项是参与伦理审批后的评估。研究可以比较从学习模型、规则倒退和控制的基线的推荐，同时衡量所感觉的有用性、选择多样性和信任。需要招聘，形式同意，撤回，安全处理数据和定义分析计划。直到这样的研究完成，界面完整性不能作为满意度的证据。

English

The fourth priority is candidate-search scaling. Exact enumeration should remain the reference while the current candidate space is bounded. If a larger catalogue makes latency unacceptable, structured pruning or approximate search should be evaluated against exact top-k recall, score loss and runtime. This would preserve a measurable accuracy-efficiency trade-off.

中文

第四个优先事项是候选搜索规模化。当前候选空间有限时，准确的编号应保持参考。如果一个更大的目录使得延迟不可接受，则结构化的剪裁或近似搜索应根据准确的顶级回忆、分数损失和运行时间进行评估。这将保持可测量的准确性-效率权衡。

English

Long-term local personalisation has a lower priority than these validation tasks. Learning from clicks can create feedback loops in which early exposure shapes later observations (Chaney et al., 2018). A future personal model would need explicit consent, reset and deletion controls, exploration, exposure logging and evaluation that separates repeated exposure from genuine preference. Local execution reduces data transfer but does not remove these methodological requirements.

中文

长期的本地定制比这些验证任务更低的优先级。从点击中学习可以创建反循环，早期暴露形成后来的观察（Chaney et al., 2018）。未来的个人模型需要明确的同意，重置和删除控制，探索，记录和评估，以分离重复暴露的真实偏好。本地执行减少了数据传输，但不会消除这些方法要求。

6.11 Summary

6.11 总结

English

The findings answer the primary question to a meaningful but restricted extent. A lightweight relational head on frozen SigLIP embeddings produced stronger offline discrimination than the controlled alternatives, and the pairwise component clearly contributed to completion ranking. Exact constrained search improved completeness according to the model score at acceptable latency for the measured candidate space. Confidence is highest for these within-data and bounded-system claims. It is lower for unsupported garment structures, menswear, new image domains and all claims involving human preference. The project's value lies in the controlled evidence and the clear boundary between what was implemented, what was measured and what remains untested.

中文

结果对主要问题有意义但有限制的答案。冻结的SigLIP嵌入式的轻量关系头产生了比控制的替代品更强的离线歧视，并且对式组件显然有助于完成排名。精确的限制搜索根据测量候选空间的模型得分在可接受的延迟下改善了完整性。这些数据内部和有限系统的要求的信心最高。对于不支持的服装结构，男性服装，新图像域和所有涉及人类偏好的要求，这种信心较低。项目价值在于可控制的证据以及实施，测量和未经测试的内容之间的清晰界限。

Chapter 7: Conclusion

第7章 结论

English

This dissertation investigated whether a lightweight outfit-compatibility model, built on frozen vision-language embeddings and combined with constrained search over abstract garment states, could support accurate, practical and auditable complete-outfit recommendation using secondary data. The result is a working research prototype and a controlled body of offline evidence. It is not a universal fashion model, a complete reproduction of published systems or a validated measure of human taste.

中文

本论文调查了基于冷视觉-语言嵌入式的轻量级穿搭兼容性模型以及抽象服装状态的有限搜索是否可以支持使用次要数据的准确、实用和可审核的完整穿搭推荐。结果是一个工作研究原型和一个受控的离线证据。它不是一个普遍的时尚模型，是发表系统的完整复制或人类品味的验证衡量。

English

Within the processed Polyvore disjoint setting, the selected model achieved test AUC of 0.8583 and FITB accuracy of 0.6025 on the retained four-position subset. Across three seeds, its mean AUC of 0.8596 exceeded the controlled recurrent, type-aware and attention-based alternatives. Structural ablation identified the pairwise module as the clearest useful component: removing it reduced mean FITB accuracy by 10.28 percentage points. The effect of clothing-position embeddings remained inconclusive because its paired interval crossed zero.

中文

在加工的聚体分离设置中，选择的模型在保留的四位子组上达到0.8583的测试AUC和0.6025的FITB精度。在三个种子中，其平均AUC为0.8596超过了控制的反复，型号意识和关注的替代方案。结构式除确定了双向模块作为最明显的有用组件：它将平均FITB精度降低了10.28个百分点。服装位置嵌入的效果仍然不确定，因为其对接间隔超过了零。

English

The model generalised consistently from validation to test, but the larger train-test gap showed that overfitting remained present. Early stopping selected the epoch-six checkpoint, temperature scaling improved validation calibration, and an outfit-length weighting alternative was rejected because it weakened the lowest-performing group and increased the group gap. These decisions illustrate a central methodological conclusion of the project: model selection should depend on several aligned measures and predefined acceptance conditions, not on one favourable aggregate score.

中文

模型从验证到测试一直保持一致，但较大的训练测试差距显示过度拟合仍然存在。早期停止选择了第六个时代的检查点，温度扩展改善了验证校准，而一个服装长度的权重替代方案被拒绝，因为它削弱了最低性能的组和增加了组差距。这些决定说明了项目的核心方法论：模型选择应取决于多个协调的措施和预定的接受条件，而不是一个有利的总分数。

English

The system contribution extended beyond classification. A training-derived artefact represented 1,670 product-independent garment concepts rather than restricting recommendations to Zara products. Exact constrained search recovered solutions that the quota-limited path frequently missed, with a recorded median CPU latency of 0.474 seconds. Weather, clothing-range, fixed-item and position constraints remained explicit, and a deterministic fallback supported operation when learned artefacts were unavailable. This separation among learned compatibility, feasible candidate generation and interface state made the system easier to inspect and test.

中文

系统贡献超出了分类。一个从训练中衍生的文物代表了1670个产品独立的服装概念，而不是限制Zara产品的推荐。精确的限制搜索恢复了数量限制的路径经常错过的解决方案，记录的中位数CPU延迟为0.474秒。天气，服装范围，固定物品和位置的限制仍然明确，并且学习文物不可用时，确定性倒退支持操作。学习兼容性，可行的候选生成和接口状态之间的分离使系统更容易检查和测试。（原文数值：1,670。）

English

The primary research question can therefore be answered as follows: the lightweight approach provides useful and comparatively strong offline compatibility ranking, to-gether with practical exact search for the present bounded candidate space, but only for the garment structures and data distributions that were evaluated. Confidence is reduced by 29.50% FITB coverage, sparse menswear evidence, the strong Polyvore- Zara domain distinction, the exclusion of one-piece garments and the absence of participant evaluation. No conclusion is drawn about user satisfaction, demographic fairness, retail conversion or universal style quality.

中文

因此，主要研究问题可以如下回答：轻量化方法提供了有用且相对强大的离线兼容性排名，与现有有限候选空间的实际精确搜索相结合，但仅对评估的服装结构和数据分布提供了实用答案。信心减少了29.50%的FITB覆盖率，男性服装的稀缺证据，强大的Polyvore-Zara域区别，排斥单件服装和缺乏参与者评估。没有关于用户满意度，人口公平性，零售转换或普遍风格质量的结论。

English

The seven objectives were addressed sufficiently to complete the stated investigation. The literature review, secondary-data pipeline, lightweight model, controlled baselines, multifaceted evaluation and integrated prototype were delivered. The objectives relating to bias and practical benefit were achieved only as critical audits: demographic performance and real user benefit remain unmeasured. This distinction is important because unavailable evidence should not be replaced by speculation.

中文

为了完成所述调查，七个目标得到了充分的解决。文献综述，二次数据管道，轻量化模型，控制基线，多方面的评估和集成原型得到了实现。关于偏见和实用效益的目标仅通过临界审计实现：人口表现和实际用户效益仍然未测量。这种区分是重要的，因为不可用证据不应被投机取代。

English

Several results changed the direction of the project rather than merely confirming its initial design. The pairwise ablation provided a stronger reason to retain relational comparison than the position-embedding experiment, whose effect remained uncertain. The overfitting audit showed why the highest training result was not an appropriate checkpoint. The length-weighting study prevented a marginal aggregate gain from being selected at the expense of the weakest observed group. The search experiment replaced an assumed efficiency benefit with a measured trade-off between latency and recovery. These outcomes demonstrate evaluation-led development: components were retained, rejected or limited according to evidence rather than because they were present in an earlier prototype.

中文

几项结果改变了项目方向,而不是仅仅确认了最初的设计。对式缩提供了比较于位置嵌入实验的关系比较强的理由,其效果仍然不确定。过度审计表明,为什么最高的训练结果不是合适的检查点。长度权重研究阻止了选择最弱的观察组的损失。搜索实验取代了假设的效率效益,测量了延迟和恢复之间的权衡。这些结果表明了基于评估的发展:组件根据证据保留,拒绝或限制,而不是因为它们存在于早期原型中。

English

The negative findings are equally relevant to the final assessment. The position embedding cannot be claimed as empirically necessary, the current data do not support a dependable comparison between clothing ranges, and the domain diagnostic warns against assuming unchanged transfer to contemporary catalogue images. The interface and functional tests address implementation quality, not human acceptance. Stating these limits narrows the claim, but also makes the positive evidence easier to interpret.

中文

负面发现同样相关于最终评估。位置嵌入不能被声称是经验必要的,目前的数据不支持服装范围之间的可靠比较,域诊断警告不要假设将不变转移到当代目录图像。接口和功能测试涉及实施质量,而不是人类接受。指出这些限制缩小了声称,但也使积极的证据更容易解释。

English

The most valuable next step is a licensed evaluation base with broader clothing-range representation, explicit one-piece structures and contemporary target-domain outfits. Repeating the controlled evaluation would test whether the relational design transfers. A participant study could follow ethics approval. Approximate search and consent-based local preference learning should be introduced only when catalogue scale and validated user needs justify them. Overall, the project contributes an evidence-backed and auditable combination of a frozen representation, focused relational model, controlled comparisons, exact search and explicit constraints.

中文

最有价值的下一步是一个拥有许可的评估基础,具有更广泛的服装范围的代表性,明确的单件结构和当代目标领域的服装。重复控制的评估将测试关系设计是否转移。参与者研究可能会遵循伦理认可。在目录规模和验证时才应引入基于同意的近似搜索和本地偏好学习。总体而言,项目贡献的结合是支持证据的,可审核的结合,包括结的代表性,集中关系模型,控制的比较,精确搜索和明确的限制。

English

Chapter A: Evidence of Required Ethics Training [Insert the student's evidence of completing all required ethics training.]

中文

第十章:要求道德培训的证据 [插入学生完成所有要求道德培训的证据。]

English

Chapter B: Confirmation of Ethical Approval Waiver [Insert the confirmation email showing that ethical approval was waived.]

中文

十

English

Chapter C: Supplementary Experimental Results C.1 Clothing-range coverage in the retained FITB subset The retained official FITB subset contained 46 menswear-labelled questions and 4,422 womenswear-labelled questions. Descriptive accuracy was 0.4348 for the menswear subset and 0.6043 for the womenswear subset. The menswear result is not treated as a reliable group comparison because its sample is substantially smaller.

中文

(原文数值: 0.4348、0.6043、1、4,422、46。)

English

Chapter D: Supplementary Software-Design Artefacts

This appendix documents the implemented prototype at a level below the methodology diagrams. Cove does not contain accounts, administrator roles, repository interfaces or relational database tables. The views therefore use UML conventions where they match the code and identify the persistent entities as a logical file schema.

中文

English

D.1 Use cases Figure D.1 records the actions available to the single local wardrobe user and the external weather dependency. Saving and wardrobe management remain local actions; the weather service receives a city query but does not receive wardrobe images.

中文

(原文数值: 1。)

English

Cove system boundary Open-Meteo geocoding Setorrestorecity and forecast service Analyse type, «include» colour and style Upload and circle garment «include» Receive complete outfit recommendation user Choose clothing range and style «extend» Inspect outfit and positional alternatives Save a piece or outfit Manage local wardrobe Figure D.1: UML use-case view of the implemented Cove prototype D.2 Overall software structure The application is predominantly function-oriented, so an overall class diagram would invent classes that do not exist. Figure D.2 uses a UML-style component view to show the real source modules, external service and stored artefacts.

中文

(原文数值: 1、2。)

English

Streamlit presentation and session-state orchestration app.py Garmentre- Recommendation Colour analysis Weather adapter Wardrobe service cognition orchestration colour_detection.py weather.py wardrobe.py recognition.py recommendation.py Figure D.2: Overall UML-style component view of the Cove software D.3 Recommendation-module classes and dependencies Figure D.3 focuses on the classes that are present in the recommendation path and includes the function-based adapter and facade on which they depend.

中文

(原文数值: 2、3。)

English

D.4 Logical entity relationships Figure D.4 expresses the multiplicities in the local wardrobe schema. It is an entity-relationship view of serialised application state, not evidence of a relational database implementation.

中文

（原文数值：4。）

English

Figure D.3: Module-level UML class and dependency diagram for recommendation Figure D.4: Entity-relationship view of the logical file schema; these entities are serialised to JSON rather than relational tables D.5 User stories and acceptance criteria Table D.1 converts the implemented interaction paths into testable user stories. Pri-orities distinguish the core recommendation journey from supporting convenience features.

中文

（原文数值：1、3、4、5。）

English

Table D.1: Implemented user stories and acceptance criteria

中文

（原文数值：1。）

Bilingual guide to figures and tables

图表双语说明

1. Functional scope of the Cove system

Cove 系统功能范围图

English

The functional-scope diagram defines what the implemented prototype accepts, processes and returns. Its left-hand column contains user and context inputs: a selected garment, city weather, style, clothing range and locally saved wardrobe content. The centre column separates garment recognition, candidate generation, complete-outfit compatibility ranking and wardrobe management. The right-hand column records the visible outcomes, including the main recommendation, a diverse alternative, model-ranked colour alternatives, weather-aware cues and saved outfits.

中文

功能范围图界定了当前原型接收什么、处理什么以及返回什么。左列是用户和情境输入，包括选中的服装、城市天气、风格、服装范围以及保存在本地的衣橱内容；中间一列把服装识别、候选生成、完整穿搭兼容性排序和衣橱管理分开；右列记录用户能够看到的结果，包括主推荐、差异化备选、由模型排序的颜色替代方案、天气提示和保存的穿搭。

English

The figure is important because it prevents the dissertation from attributing every function to the learned model. The model scores complete candidate outfits, whereas recognition, weather validation, storage and explicit feasibility rules are separate system responsibilities. The arrows show the main dependencies rather than every user-interface event.

中文

这张图的重要意义在于，它避免论文把所有功能都归因于学习模型。模型负责给完整候选穿搭评分，而识别、天气校验、存储和明确的可行性规则属于其他系统模块。图中的箭头表示主要依赖关系，而不是穷举每一个界面事件。

2. Technical route

技术路线图

English

The technical-route diagram follows the research from secondary fashion data to the evaluated Cove prototype. Polyvore disjoint images and metadata are mapped to four supported clothing positions and used to construct positive and corrupted negative outfits. A frozen SigLIP encoder converts garments into 768-dimensional representations. These processed tensors are then used to train the compatibility head, select a checkpoint with validation data and calibrate its output.

中文

技术路线图从二手时尚数据开始，一直追踪到经过评价的 Cove 原型。Polyvore disjoint 的图片和元数据首先被映射到四个支持的服装位置，并用于构造正穿搭和人工破坏的负穿搭；冻结的 SigLIP 编码器把服装转换成 768 维表示；处理后的张量用于训练兼容性模型头、依据验证集选择检查点，并对输出进行校准。

English

The lower part of the diagram connects model development to baselines, FITB evaluation, ablation, bias and generalisation checks. The selected checkpoint and abstract clothing prototypes then enter constraint filtering and exact candidate search. This figure functions as a concise map of the methodology and evidence chain; it also makes clear that Zara is not the source of compatibility labels or the only recommendation catalogue.

中文

图的下半部分把模型开发与基线比较、FITB 评价、消融、偏差和泛化检查连接起来。所选模型检查点与抽象服装原型随后进入约束过滤和精确候选搜索。这张图相当于方法与证据链的简明地图，同时明确说明 Zara 既不是兼容性标签来源，也不是唯一的推荐候选库。

3. Lightweight compatibility-model architecture

轻量兼容性模型架构图

English

The architecture diagram shows the trainable task-specific part of the recommendation model. Up to four frozen 768-dimensional SigLIP embeddings and a position mask enter layer normalisation, linear projection and GELU activation. Learned position embeddings are concatenated to the projected item representations. The model then forms a masked global outfit summary and six absolute pairwise-difference representations, corresponding to all pairs among four positions.

中文

模型架构图展示推荐模型中针对本任务训练的部分。最多四个冻结的 768 维 SigLIP 向量和一个位置掩码首先经过层归一化、线性投影与 GELU 激活；学习得到的位置嵌入随后与投影后的单品表示拼接。模型再形成带掩码的全局穿搭摘要，并构造四个位置之间全部六组绝对成对差异表示。

English

The global and pairwise summaries are concatenated and passed through a small multilayer perceptron to produce a compatibility logit, which is converted into a calibrated score for runtime ranking. The pairwise branch is central because compatibility concerns relationships between garments rather than only the average appearance of an outfit. The diagram also communicates the project's lightweight boundary: SigLIP representation learning is reused and frozen; only the smaller compatibility head is trained.

中文

全局摘要与成对摘要拼接后进入小型多层感知机，产生兼容性 `logit`，并转换成运行时排序使用的校准分数。成对关系分支很关键，因为穿搭兼容性关注衣服之间的关系，而不只是整套衣服外观的平均值。该图还说明了项目的轻量化边界：SigLIP 的表示学习成果被复用并冻结，真正训练的是较小的兼容性模型头。

4. Evaluation framework

评价框架图

English

The evaluation-framework diagram begins with disjoint training, validation and held-out test data and applies the same data protocol to the selected model and controlled alternatives. Evidence is divided into four groups: binary discrimination, completion ranking, robustness and complete-system behaviour. Corresponding measures include AUC and threshold metrics; FITB, MRR and NDCG; multiple seeds, ablation and calibration; and coverage, exact search and runtime.

中文

评价框架图从相互独立的训练集、验证集和保留测试集出发，并把一致的数据协议用于最终模型和受控替代模型。证据被划分为四组：二分类区分、穿搭补全排序、稳健性以及完整系统行为。相应指标包括 AUC 与阈值指标；FITB、MRR 和 NDCG；多随机种子、消融与校准；以及覆盖率、精确搜索和运行时间。

English

Its purpose is to show that no single score is sufficient for the dissertation's claims. Confidence intervals, retained benchmark coverage and stated limitations accompany the metrics. Model promotion and deployment are therefore based on several forms of evidence rather than the highest observed AUC alone.

中文

这张图的作用是说明，单一分数不足以支持论文结论。指标需要同时配合置信区间、保留的基准覆盖率和明确的局限解释。因此，模型晋级和系统部署依据多种证据，而不是只选择观察到的最高 AUC。

5. Runtime recommendation and fallback flow

运行时推荐与回退流程图

English

The runtime-flow diagram starts when a user selects a garment from a photograph or the local wardrobe. The system analyses its broad type, colour and style and obtains a SigLIP representation. It then checks whether the compatible model checkpoint and candidate artefacts are available. When they are available, the selected item is fixed, infeasible states are filtered by position, weather, style and clothing range, and complete candidates are scored in batches through exact enumeration.

中文

运行时流程图从用户在照片或本地衣橱中选一件服装开始。系统分析它的大类、颜色和风格，并取得 SigLIP 表示；随后检查兼容性模型检查点和候选产物是否可用。如果这些文件可用，系统就固定用户选择的服装，根据位置、天气、风格和服装范围过滤不可行状态，再通过精确枚举分批评价完整候选组合。

English

If learned artefacts are unavailable or incompatible, the system follows a deterministic content-based fallback rather than stopping. Both paths produce a primary outfit, a different alternative and feasible position-level replacements. This flow documents operational resilience while keeping the fallback claim bounded: continued operation does not mean that a rule-ranked output is equivalent to a learned compatibility score.

中文

如果学习模型产物缺失或不兼容，系统会使用确定性的内容规则回退，而不是直接停止。两条路径最终都生成主穿搭、具有明显差异的备选，以及可行的位置级替代方案。该流程体现了系统的运行韧性，但同时限制了结论边界：系统能够继续运行，不代表规则排序结果与学习得到的兼容性分数等价。

6. Logical local-data relationships

本地数据逻辑关系图

English

This diagram presents how user state and research artefacts relate at runtime. ``wardrobe.json`` contains arrays of pieces, outfits and outfit groups. A saved piece may reference a local PNG file, while a saved outfit contains values or absences for four supported positions. ``user_preferences.json`` stores up to six saved cities. The application loads these local records together with the compatibility checkpoint and abstract prototype catalogue.

中文

这张图展示用户状态与研究产物在运行时如何关联。``wardrobe.json`` 包含单品、穿搭和穿搭分组数组；保存的单品可以引用本地 PNG 文件，保存的穿搭则为四个支持位置保存服装值或缺失状态；``user_preferences.json`` 最多保存六个城市。应用运行时会同时载入这些本地记录、兼容性模型检查点以及抽象服装原型目录。

English

Evaluation JSON files, manifests, hashes and training histories form a separate research-evidence layer and are not user profiles. The figure deliberately describes logical relationships instead of database tables because the prototype does not implement a relational database. Its value is accuracy and auditability: readers can see which information is personal, which files support inference and which outputs document the experiments.

中文

评价 JSON、清单、哈希和训练历史构成单独的研究证据层，并不属于用户画像。由于原型没有实现关系型数据库，这张图有意描述逻辑关系，而不是虚构数据库表。它的价值在于准确性和可审计性：读者能够分辨哪些是个人信息、哪些文件支持模型推理，以及哪些输出用于记录实验。

7. UML use-case view

UML 用例图

English

The use-case diagram records the interactions available to a single local wardrobe user. The user can set or restore a city, upload and circle a garment, choose style and clothing range, request a complete outfit, inspect alternatives, save results and manage the wardrobe. Garment analysis is included by image selection and recommendation. Inspecting alternatives extends the main recommendation journey rather than being required for every result.

中文

用例图记录单个本地衣橱用户可以进行的交互。用户能够设置或恢复城市、上传并圈选服装、选择风格与服装范围、请求完整穿搭、查看备选、保存结果以及管理衣橱。服装分析是图片选择和推荐流程所包含的步骤；查看备选则是对主推荐流程的扩展，并非每一次结果都必须执行。

English

Open-Meteo appears outside the Cove system boundary as the only external service in this view. It receives a city query for geocoding and weather retrieval but does not receive wardrobe images. The diagram is useful for defining actors, privacy boundaries and implemented user goals without introducing administrator roles or login flows that do not exist in the prototype.

中文

Open-Meteo 位于 Cove 系统边界之外，是该图中的外部服务。它只接收城市查询以进行地理编码和天气获取，不接收衣橱图片。这张图用于明确参与者、隐私边界和已经实现的用户目标，同时避免加入原型中并不存在的管理员角色或登录流程。

8. Overall UML-style component view

整体 UML 风格组件图

English

The overall component diagram places the Streamlit presentation and session-state orchestration at the top. It calls garment recognition, colour analysis, weather, recommendation and wardrobe modules. The recommendation module can use either the trained-model adapter and candidate ranker or the deterministic fallback. Separate stores provide the model checkpoint, abstract prototypes, local JSON documents and wardrobe images, while Open-Meteo supplies weather data.

中文

整体组件图把 Streamlit 展示层和会话状态编排放在顶端。它调用服装识别、颜色分析、天气、推荐和衣橱模块。推荐模块既可以经过训练模型适配器和候选排序器，也可以进入确定性回退。模型检查点、抽象原型、本地 JSON 文档和衣橱图片由不同存储提供，天气数据则来自 Open-Meteo。

English

This is a component view rather than a conventional overall class diagram because most of the application is function-oriented Python. Inventing service, repository and data-transfer-object classes would misrepresent the code. The diagram's function is to communicate real software boundaries and dependencies at a readable scale.

中文

这是一张组件图，而不是传统意义上的整体类图，因为应用的大部分 Python 代码采用函数式组织。如果凭空加入 Service、Repository 或 DTO 类，就会错误描述实际代码。该图的功能是在容易阅读的尺度上表达真实的软件边界与依赖关系。

9. Recommendation-module class and dependency view

推荐模块类与依赖图

English

This module-level diagram focuses on the object-oriented structures that genuinely exist in the recommendation path. `ModelConfig` supplies the embedding, hidden, position and dropout settings used to construct `CompatibilityRanker`. The model exposes forward scoring, preprojected scoring and probability conversion. `OutfitCandidateRanker` holds the trained model and performs exact or bounded candidate ranking while recording search statistics.

中文

这张模块级图只聚焦推荐路径中确实存在的面向对象结构。`ModelConfig` 提供构造 `CompatibilityRanker` 所需的向量维度、隐藏层维度、位置维度和 dropout 设置；模型提供前向评分、预投影评分和概率转换；`OutfitCandidateRanker` 持有训练模型，执行精确或有限候选排序，并记录搜索统计。

English

The lower row adds the function-based trained-model adapter and recommendation facade, plus the deterministic `LightweightOutfitRanker`. Solid arrows indicate construction or ownership relationships; dashed arrows indicate software dependencies. Its purpose is to connect the architecture described in the methodology to identifiable code units and to show where learned and fallback paths diverge.

中文

下方一行补充了函数式训练模型适配器、推荐门面以及确定性的 `LightweightOutfitRanker`。实线箭头表示构造或持有关系，虚线箭头表示软件依赖。该图的意义在于把方法章节描述的架构对应到可以识别的代码单元，并明确学习路径与回退路径在哪里分开。

10. Logical entity-relationship view

逻辑实体关系图

English

The entity-relationship view describes the logical schema serialised to local files. One wardrobe store can contain zero or more pieces and outfits and at least the default outfit groups. Each outfit contains exactly four slot entries; each slot contains either a snapshot of one piece or a null value. A piece may reference one image file, and a group may contain multiple outfits. Saved city preferences are stored separately.

中文

实体关系图描述序列化到本地文件中的逻辑模式。一个衣橱存储可以包含零个或多个单品与穿搭，并至少包含默认穿搭分组；每套穿搭固定含有四个位置条目，每个位置保存一件服装的快照或空值；一件单品可以引用一个图片文件，一个分组可以包含多套穿搭；收藏城市则单独保存。

English

Cardinalities make behaviours such as partial outfits, piece reuse and grouping explicit. The figure must not be interpreted as evidence of SQL tables, primary keys or foreign-key enforcement. Its function is to document persistence rules and support reasoning about deletion, reuse and privacy in the implemented file-based design.

中文

图中的基数关系明确表达了未完成穿搭、单品复用和分组等行为。不能把这张图理解为系统存在 SQL 表、主键或外键约束的证据。它的功能是记录持久化规则，并支持对当前文件式设计中的删除、复用和隐私问题进行分析。

11. Core application-functions table

核心应用功能表

English

The core-functions table assigns a principal input, output and responsibility to garment analysis, candidate control, compatibility ranking, wardrobe management and fallback. It is a compact responsibility matrix rather than an experiment table. The separation shows that the compatibility head evaluates relational coherence only after candidates have been made feasible by explicit controls.

中文

核心功能表为服装分析、候选控制、兼容性排序、衣橱管理和回退分别指定主要输入、输出与职责。它是一张紧凑的职责矩阵，而不是实验结果表。这样的划分说明，候选首先由明确规则保证可行，之后兼容性模型头才负责评价服装之间的关系协调性。

English

This table supports traceability between the product functions and the implementation. It also makes testing easier to explain: recognition accuracy, constraint correctness, model ranking and local persistence are different concerns and should not be represented by one metric.

中文

该表支持产品功能与代码实现之间的可追踪性，也使测试设计更容易说明：识别准确性、约束正确性、模型排序和本地持久化属于不同问题，不能用同一个指标概括。

12. Local persistence and research-artefacts table

本地持久化与研究产物表

English

This table lists the concrete files behind the logical data diagrams. It distinguishes the local wardrobe store, local garment images and saved-city preferences from the compatibility checkpoint, abstract candidate prototypes and experimental evidence. Each row states the principal contents and explains how the artefact relates to the running application or the research audit trail.

中文

这张表列出逻辑数据图背后的具体文件，把本地衣橱记录、用户服装图片和收藏城市偏好，与兼容性模型检查点、抽象候选原型及实验依据区分开来。每一行都说明文件的主要内容，以及它与运行中的应用或研究审计链之间的关系。

English

The distinction is meaningful for privacy and reproducibility. User files remain local and are Git-ignored, whereas model and evaluation artefacts document which system version produced the reported results. The table therefore complements the relationship diagrams with implementation-level file names.

中文

这种区分对于隐私和可复现性很重要。用户文件保留在本地并被 Git 忽略；模型和评价产物则记录哪一个系统版本产生了论文中报告的结果。因此，该表用实现层面的文件名补充了关系图。

13. User stories and acceptance criteria

用户故事与验收标准表

English

The user-story table translates twelve implemented interaction paths into testable requirements. The stories cover location and saved cities, image selection and recognition, context controls, outfit completion, alternatives, fallback, saving, editing, the reusable piece library and local persistence. Each story contains an observable acceptance criterion and is assigned a Must or Should priority.

中文

用户故事表把十二条已实现的交互路径转换成可测试需求，覆盖地点与收藏城市、图片选择与识别、情境控制、穿搭补全、备选、回退、保存、编辑、可复用单品库和本地持久化。每条故事都有可以观察的验收条件，并被标记为 Must 或 Should 优先级。

English

The table's purpose is requirements traceability rather than claiming that a formal industrial agile process was followed. It helps a reader check whether the implementation meets the intended user journey and identifies which supporting conveniences are less critical than the core recommendation and privacy requirements.

中文

该表的目的在于需求可追踪性，而不是声称项目完整采用了正式工业敏捷流程。它帮助读者检查实现是否覆盖预期用户旅程，并区分哪些便利功能的优先级低于核心推荐与隐私要求。

14. How the diagrams work together

图表之间如何配合

English

The figures operate at different levels and should not be read as duplicates. The functional-scope and use-case views describe what the user can do. The technical route and evaluation framework describe how the research was conducted and judged. The model-architecture and recommendation-class views explain the learned scorer and its code dependencies. The overall component and runtime-flow views describe deployment behaviour. Finally, the local-data, ER and persistence views document storage and privacy boundaries.

中文

这些图位于不同层次，不应被视为重复内容。功能范围图和用例图说明用户能够做什么；技术路线图和评价框架图说明研究怎样开展、怎样判断；模型架构图和推荐类图解释学习式评分器及其代码依赖；整体组件图和运行时流程图描述部署行为；最后，本地数据图、ER 图和持久化表记录存储方式与隐私边界。

English

Together they support three dissertation needs: communicating the product clearly, demonstrating that the software was actually engineered, and connecting implementation decisions to research evidence. They do not prove recommendation quality by themselves; quality claims remain grounded in the reported experiments, confidence intervals, coverage and limitations.

中文

这些图表共同支持论文的三项需要：清楚介绍产品、证明软件确实经过工程实现，以及把实现决策与研究证据连接起来。图表本身不能证明推荐质量；关于质量的结论仍必须建立在论文报告的实验、置信区间、覆盖率和局限之上。

English

During review, check whether every label matches the current code, whether any arrow implies a function the system does not perform, and whether the distinction between learned ranking and explicit constraints remains clear. The most important conceptual boundary is that the model supplies a compatibility score, while the full recommender also depends on recognition, candidate construction, weather and clothing-range rules, exact search, storage and fallback.

中文

通读时应重点检查：每个标签是否与当前代码一致、是否有箭头暗示系统并未执行的功能，以及学习式排序与明确约束之间的区别是否清楚。最重要的概念边界是：模型只提供兼容性分数，而完整推荐系统还依赖识别、候选构造、天气与服装范围规则、精确搜索、存储和回退机制。

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Bibliographic titles, journal names, DOIs and URLs are preserved in their published form to avoid altering source information.

为避免改变文献来源信息，题名、期刊名、DOI 与链接均保留正式英文形式。

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